

# Ultrassonografia abdominal simplificada: guia passo a passo

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 [pocus101.com/abdominal-ultrasound-made-easy-step-by-step-guide](https://pocus101.com/abdominal-ultrasound-made-easy-step-by-step-guide)

Vi Dinh

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Ao se deparar com um paciente com sintomas abdominais mal localizados, o abdômen pode parecer uma caixa preta estressante, sem quase nenhuma pista para guiar o exame.

A ultrassonografia à beira do leito (POCUS) está mudando esse cenário, pois oferece uma visão detalhada do que pode estar acontecendo. Usar a POCUS para solucionar casos abdominais ajudará você a aprimorar suas habilidades de diagnóstico, otimizar sua tomada de decisões clínicas e descartar hipóteses diagnósticas com confiança – tudo isso sem precisar levar o paciente para a tomografia computadorizada ou sala de radiologia!

Neste artigo, usaremos uma abordagem baseada em órgãos para ensinar você a realizar exames de ultrassom abdominal completos, incluindo **figado, vesícula biliar, intestino delgado e grosso, apêndice e baço** . Também explicaremos como diagnosticar diversas patologias abdominais e forneceremos uma breve visão geral de algumas aplicações pediátricas do ultrassom abdominal.

Índice

## Indicações para ultrassonografia abdominal

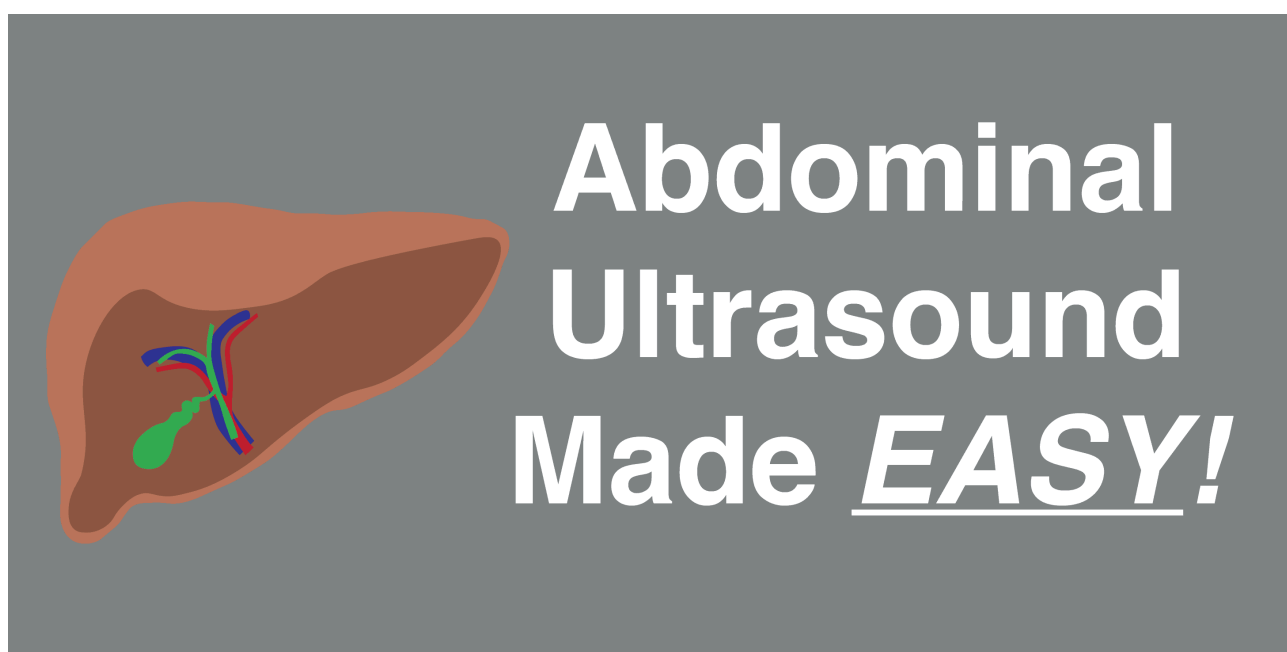
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Como a cavidade abdominal contém muitos órgãos, provavelmente é mais fácil pensar no diagnóstico diferencial em termos de quadrantes/áreas abdominais. Abaixo, segue uma lista e uma figura mostrando alguns diagnósticos abdominais comuns em relação às regiões abdominais.

No entanto, lembre-se de que esses diagnósticos *não* se restringem apenas a essas regiões. Por exemplo, a diverticulite geralmente se manifesta como dor no quadrante inferior esquerdo, mas às vezes pode causar dor no quadrante inferior direito.

Esta lista e figura são apenas pontos de partida para você entender as patologias abdominais mais comuns e onde elas geralmente se manifestam.

- **Dor abdominal:**
  - *Quadrante superior direito:* coledoclitíase, hepatite, avaliação para líquido livre, cálculos renais (também dor lombar)
  - *Quadrante inferior direito:* apendicite, intussuscepção, abscesso do psoas, hérnia inguinal.
  - *Quadrante superior esquerdo:* esplenomegalia, avaliação para líquido livre na cavidade renal, cálculos renais (também dor lombar).
  - *Quadrante inferior esquerdo:* diverticulite, obstrução do intestino delgado, hérnia inguinal.
  - *Epigástrico :* AAA, pancreatite, obstrução intestinal (também pode ser generalizada)
  - *Pélvico:* retenção urinária, gravidez, gravidez ectópica, massa pélvica, avaliação de líquido livre
  - *Testicular:* torção, epididimite, hidrocele.
- **Trauma abdominal :** Avaliação de líquido livre na cavidade abdominal (hemoperitônio)



*Patologia abdominal por região*

Neste artigo, abordaremos as aplicações mais comuns do ultrassom abdominal à beira do leito (POCUS), incluindo: **avaliação hepatobiliar, do baço, do apêndice, detecção de líquido livre, emergências intestinais (obstrução do intestino delgado/pneumoperitônio) e aplicações abdominais específicas para pacientes pediátricos (estenose pilórica, intussuscepção).**

Abordamos [a ultrassonografia pélvica](#) , [a ultrassonografia obstétrica](#) , [a ultrassonografia da bexiga](#) , [a ultrassonografia renal](#) e [a ultrassonografia da aorta abdominal](#) em publicações separadas (clique nos links para acessar essas publicações).

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## Preparação para ultrassom abdominal

### Preparação do Paciente

- O paciente deve estar em decúbito dorsal com a cabeceira da cama plana.
- Peça ao paciente que dobre as pernas pelos joelhos, se possível, para ajudar a relaxar os músculos abdominais.

- Algumas aplicações, como o exame de ultrassom da vesícula biliar, exigirão que o paciente esteja em decúbito lateral esquerdo.

## Preparação do aparelho de ultrassom

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- **Transdutor** : [Sonda ultrassônica curvilínea](#) (escolha ideal) ou [sonda phased array](#) .  
Pode ser necessário usar uma [sonda linear](#) para apendicite e em aplicações pediátricas.
- **Pré-configuração** : Abdômen
- **Posicionamento do aparelho** : Posicione o aparelho de ultrassom à direita do paciente, com a tela voltada para você. Nessa configuração, você poderá ficar de frente tanto para o paciente quanto para a tela do ultrassom, realizando o exame com a mão direita e manipulando os botões do aparelho com a mão esquerda.

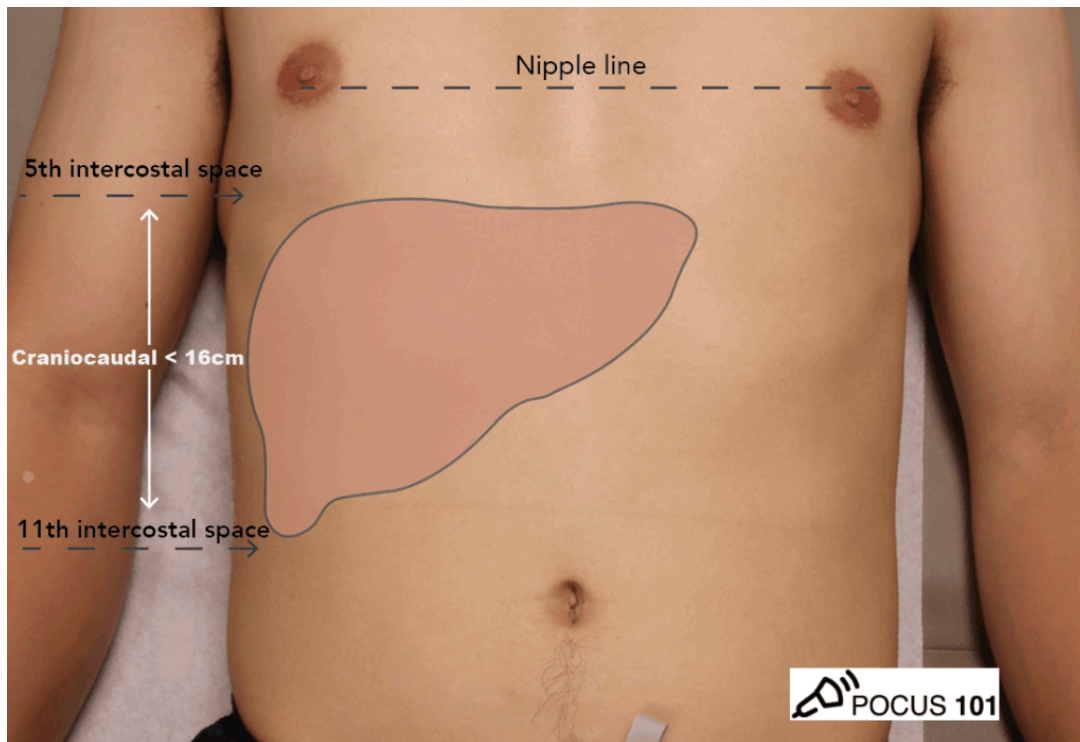


## Ultrassonografia do fígado

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Você pode utilizar o ultrassom à beira do leito do fígado quando se deparar com um paciente com dor no quadrante superior direito (QSD) e alterações agudas nos testes de função hepática.





Para os fins deste guia de ultrassom abdominal, a medida mais importante a ser considerada é o comprimento craniocaudal do lobo direito. Embora o tamanho do fígado varie consideravelmente de paciente para paciente, dependendo da idade, sexo, biotipo e outros fatores, um fígado saudável deve ter **menos de 16 cm** nessa dimensão. Isso corresponde aproximadamente ao 5º ao 11º espaço intercostal direito.

## Protocolo de ultrassom hepático

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Utilizando a abordagem ultrassonográfica lateral, é possível avaliar o tamanho, a textura (ecogenicidade do parênquima) e as características da superfície (contorno capsular) do fígado (Rumack). É importante ressaltar que, a partir dessa perspectiva, o fígado também oferece uma janela acústica útil para visualizar estruturas clinicamente importantes, como a [aorta](#) , [a veia cava inferior](#), [as veias hepáticas](#) e [a veia porta](#) .

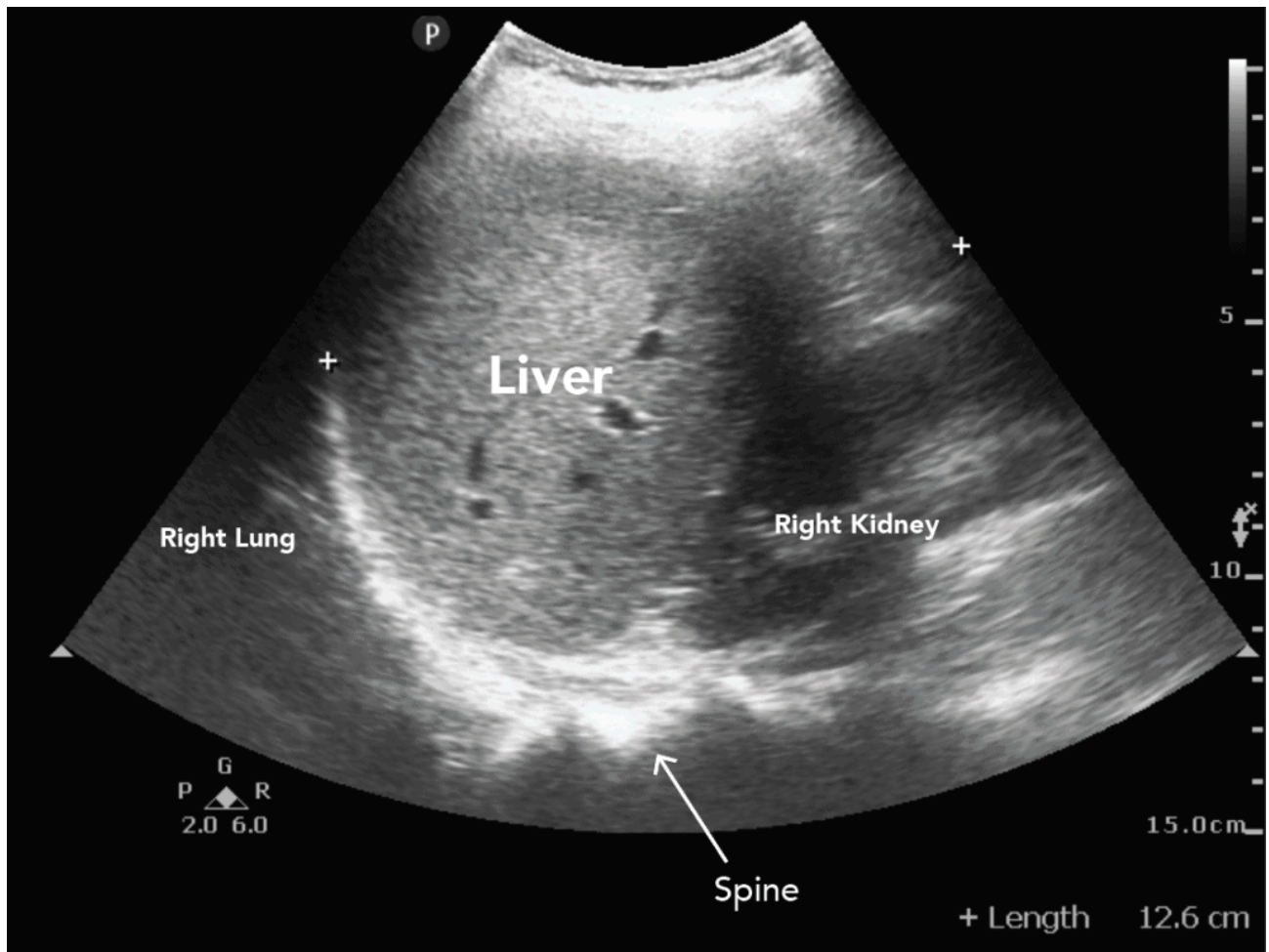
**Posicione a sonda** no lado direito do paciente, na **linha axilar média**, **entre o 10º e o 11º** espaço intercostal, com o indicador **voltado para a cabeça do paciente**. Se possível, peça ao paciente que levante o braço direito acima da cabeça para ampliar os espaços intercostais.



Posicione a sonda para visualizar o fígado. O indicador deve apontar para a cabeça do paciente (seta vermelha).

**Visualize o diafragma , o fígado e o rim direito** no eixo longitudinal.

*Dica: se as costelas estiverem obstruindo a visão, peça ao paciente para prender a respiração profundamente para ampliar ainda mais os espaços intercostais. Tente também girar a sonda obliquamente para alinhá-la com os espaços intercostais.*



*Vista lateral do fígado, com o pulmão direito acima e o rim direito abaixo.*

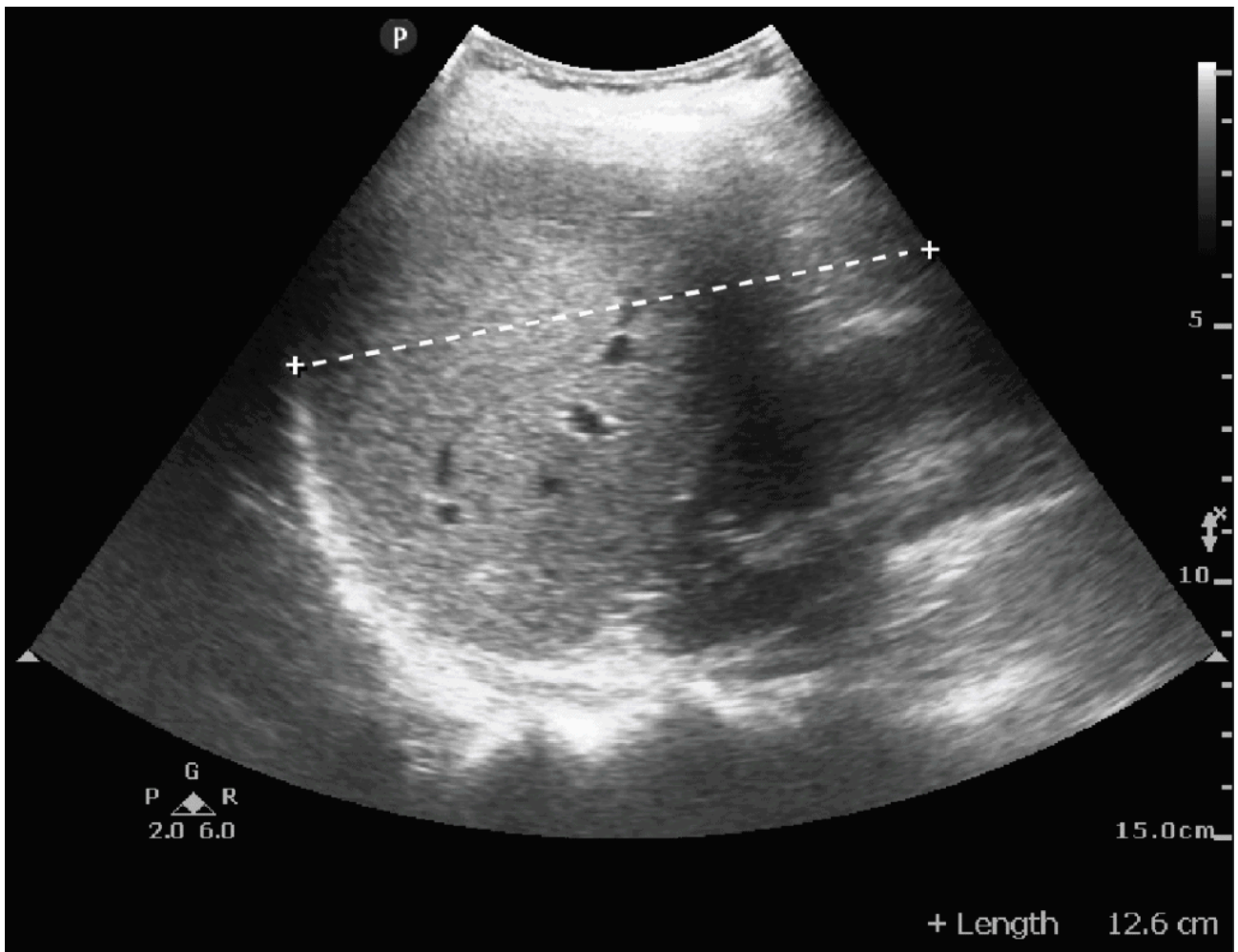
### **Avalie a ecogenicidade e o contorno capsular do fígado.**

- Normalmente, o fígado apresenta **ecogenicidade homogênea**, semelhante em brilho ao córtex renal (Rumack). Verifique a presença de variações notáveis, como aumento ou diminuição da ecogenicidade, massas ou lesões.
- O fígado deve apresentar um **contorno capsular regular**, semelhante ao do rim. Observe qualquer irregularidade ou nodularidade acentuada.

### **Measure the liver span in the craniocaudal dimension.**

Once you have visualized the structures listed above, [freeze the image](#) and [measure the liver span](#) from the diaphragmatic surface to the inferior border, as pictured below.

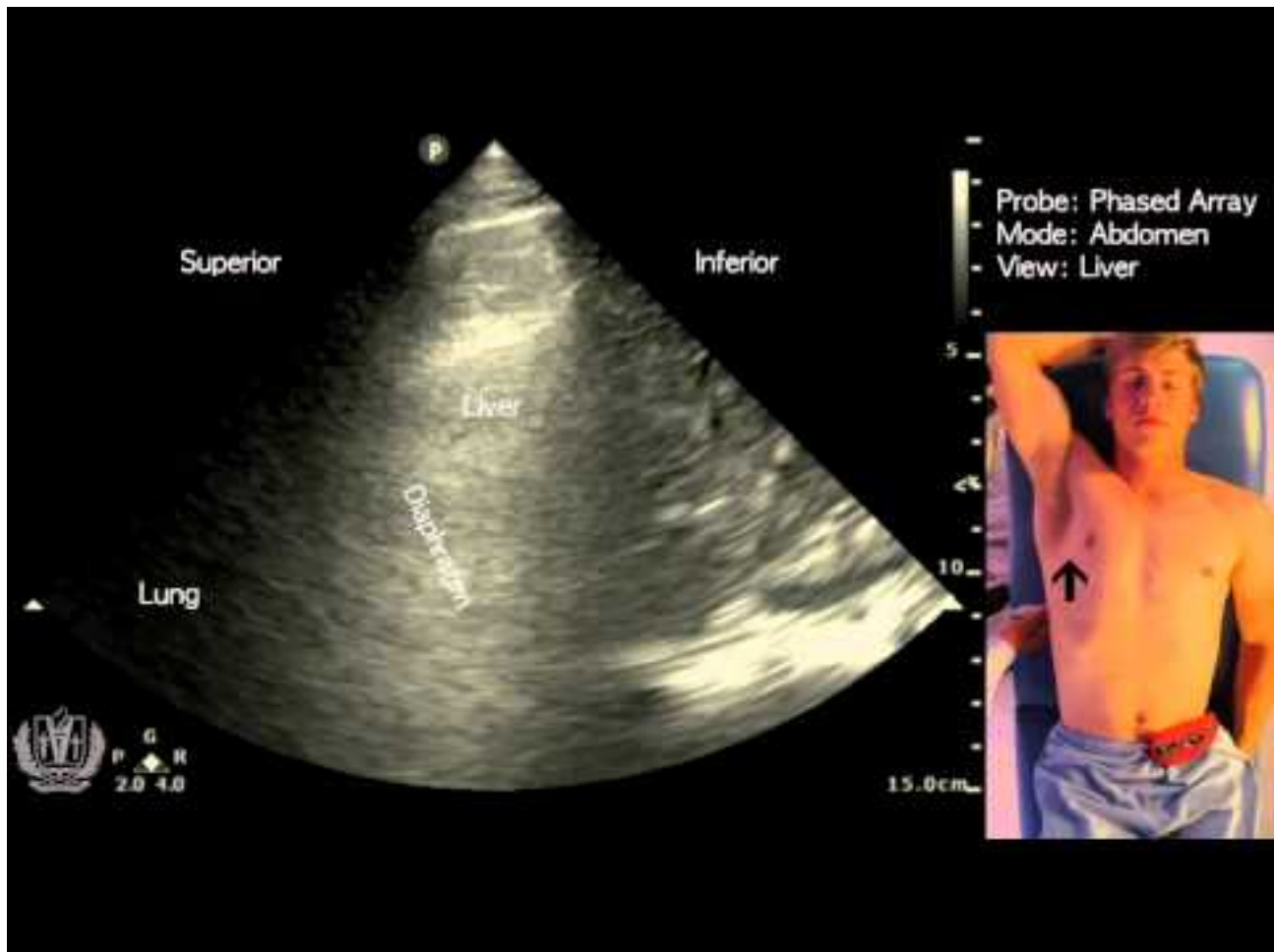
*POCUS 101 Tip: sometimes, the ultrasound window is too narrow to capture the liver's diaphragmatic surface and tip together on the screen. Estimate, to the best of your ability, where either of these surfaces would end if you could see them both, and place your calipers at your best guess.*



*This liver measures 12.6cm in the craniocaudal dimension*

Here is a quick video summarizing how to look for the liver with ultrasound:





[Watch on YouTube](#)

## Liver Pathology

Scanning the liver using ultrasound has many applications that are beyond the scope of this post. Here, we focus on the findings of acute hepatitis, cirrhosis, and fatty liver disease since they comprise the bulk of liver-related morbidity and mortality (Edlin, Scaglione).

Here is a quick table to help you differentiate between the different liver pathologies on ultrasound.

| Acute hepatitis:                                                                               | Chronic hepatitis/cirrhosis:                                                              | Fatty Liver Disease (Hepatic Steatosis)                                             |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| ↑ Brightness of portal vein walls<br>↓ The echogenicity of the liver<br>Usually Enlarged Liver | ↓ Brightness of portal vein walls<br>↑ Liver echogenicity<br>Usually Small/Atrophic Liver | ↓ Brightness of portal vein walls<br>↑ Liver echogenicity<br>Usually enlarged Liver |

However, the findings described here for hepatitis and cirrhosis are relatively uncommon findings in the general population and have uncertain operator-dependent sensitivities and

specificities. Any ultrasound exam of the liver must be accompanied by a thorough history and physical exam, and inconclusive findings must be bolstered by additional diagnostic modalities.

## Acute Hepatitis Ultrasound Findings

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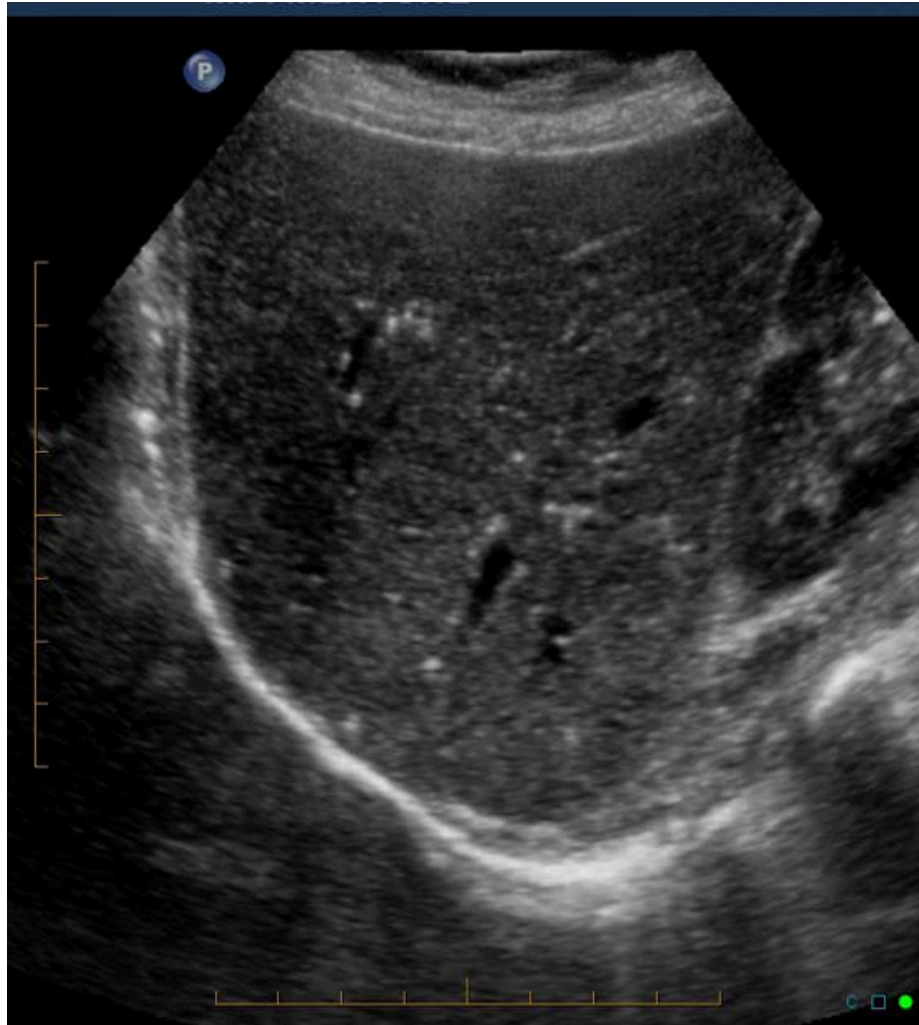
In the U.S. alone, over 3 million people have viral hepatitis (hepatitis C > hepatitis B), and over half of those people do not know they have the virus. What's more, millennials currently represent the most commonly affected population, followed closely by baby boomers. (Office of Infectious Disease).

Acute hepatitis occurs when the liver has an inflammatory reaction to an injury, whether traumatic, infectious, drug-induced, or autoimmune (Joshi). This causes the liver to become edematous and enlarged.

Patients with acute hepatitis most frequently present with acute RUQ pain, fever, jaundice, and elevated liver enzymes (AST/ALT).

**Hepatomegaly** is the most sensitive POCUS finding in acute hepatitis, defined as a **craniocaudal length  $\geq 16.0$  cm** in the midclavicular line (Tchelepi). In addition, the liver may appear relatively *hypoechoic* versus the adjacent kidney, due to inflammatory fluid buildup in acute hepatitis. This causes the parenchyma to resemble a “**starry sky**” appearance, whereby the portal vein walls appear as hyperechoic “stars” on a background of dark, edematous hepatocytes. However, this finding has low sensitivity and specificity (Heller).





Acute Hepatitis – Edematous Hypoechoic Liver with “Starry Sky” Sign

Reference: <https://radiopaedia.org/cases/acute-hepatitis-starry-sky-appearance>

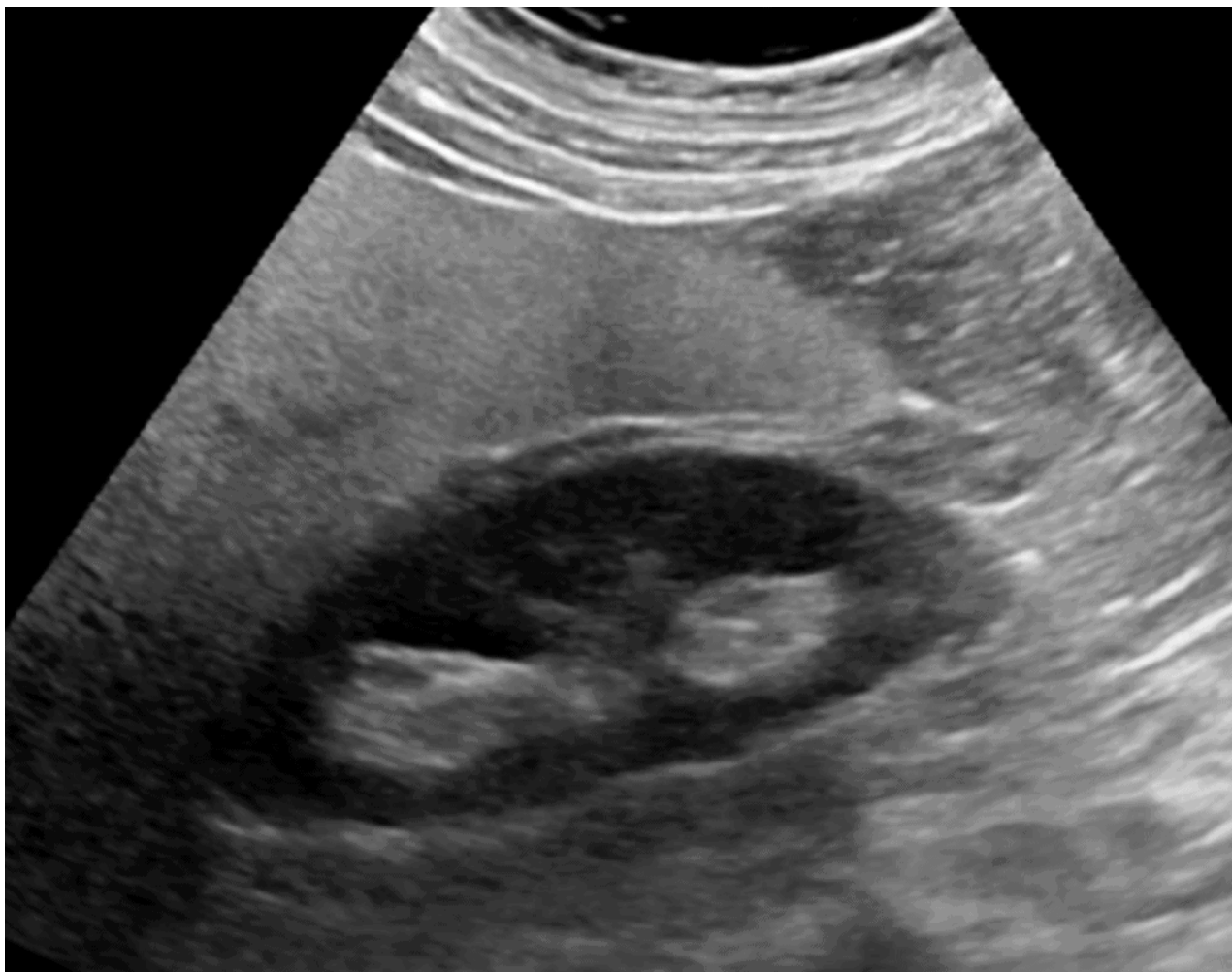
## Cirrhosis Ultrasound Findings

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Cirrhosis results from chronic damage to the liver. In the U.S., excessive alcohol consumption and chronic hepatitis C infection are the most common causes (Muzio). Over half a million Americans are estimated to have cirrhosis, although nearly 70% of them report not knowing they had liver disease. (Office of Infectious Disease).

Symptoms can include ascites, hepatosplenomegaly, jaundice, and dermatological findings such as palmar erythema or spider angiomas.

On ultrasound, you may find **surface (capsular) nodularity** (88% sensitivity, up to 95% specificity). In addition, the liver will be **hyperechoic** relative to the right kidney. Other, more difficult findings to assess include right lobe atrophy and caudate lobe hypertrophy (Arger). Keep in mind, however, that the liver may be enlarged in the early stages of cirrhosis, and you will not see atrophy until end-stage disease.



*In cirrhosis, the liver becomes markedly hyperechoic relative to the kidney.*  
Case courtesy of Dr Chris O'Donnell, [Radiopaedia.org](http://Radiopaedia.org). From the case [rID: 18858](#)

## Fatty Liver Disease (Hepatic Steatosis)

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Fatty liver disease or Hepatic Steatosis is a common cause of abdominal pain and elevated liver function test enzymes.

Fatty liver disease can be associated with alcohol abuse, diabetes, obesity, hyperlipidemia, medications (amiodarone, methotrexate, chemotherapy), pregnancy, metabolic disorders, etc.

### **Hepatic Steatosis ultrasound findings include (Khov 2014):**

- Increased liver echogenicity (fat appears echogenic on ultrasound).
- May have enlarged liver size.
- The Renal cortex appears relatively hypoechoic when compared to the liver (the kidney cortex and liver are usually isoechoic).



Fatty Liver Disease (hepatic steatosis) on Ultrasound. (Image from Radiopaedia.org)

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## Gallbladder & Common Bile Duct Ultrasound

Ultrasound is the imaging modality of choice for most biliary tree pathologies. However, since the gallbladder is not fixed to the body wall like other GI organs, it can have a variety of positions in the right upper quadrant (RUQ), so getting a good view can be tricky. Here, we provide a way of imaging this somewhat challenging organ.

Ideally, your patient should fast prior to the exam, so the gallbladder will be dilated with anechoic bile. However, you will most likely have to image the gallbladder in an emergent setting, and fasting will not have been possible.

### Gallbladder Ultrasound Protocol

Make sure the patient is fasting. Eating food will make the gallbladder contracted and you may not be able to visualize it. Tip: if possible, have your patient assume the left lateral

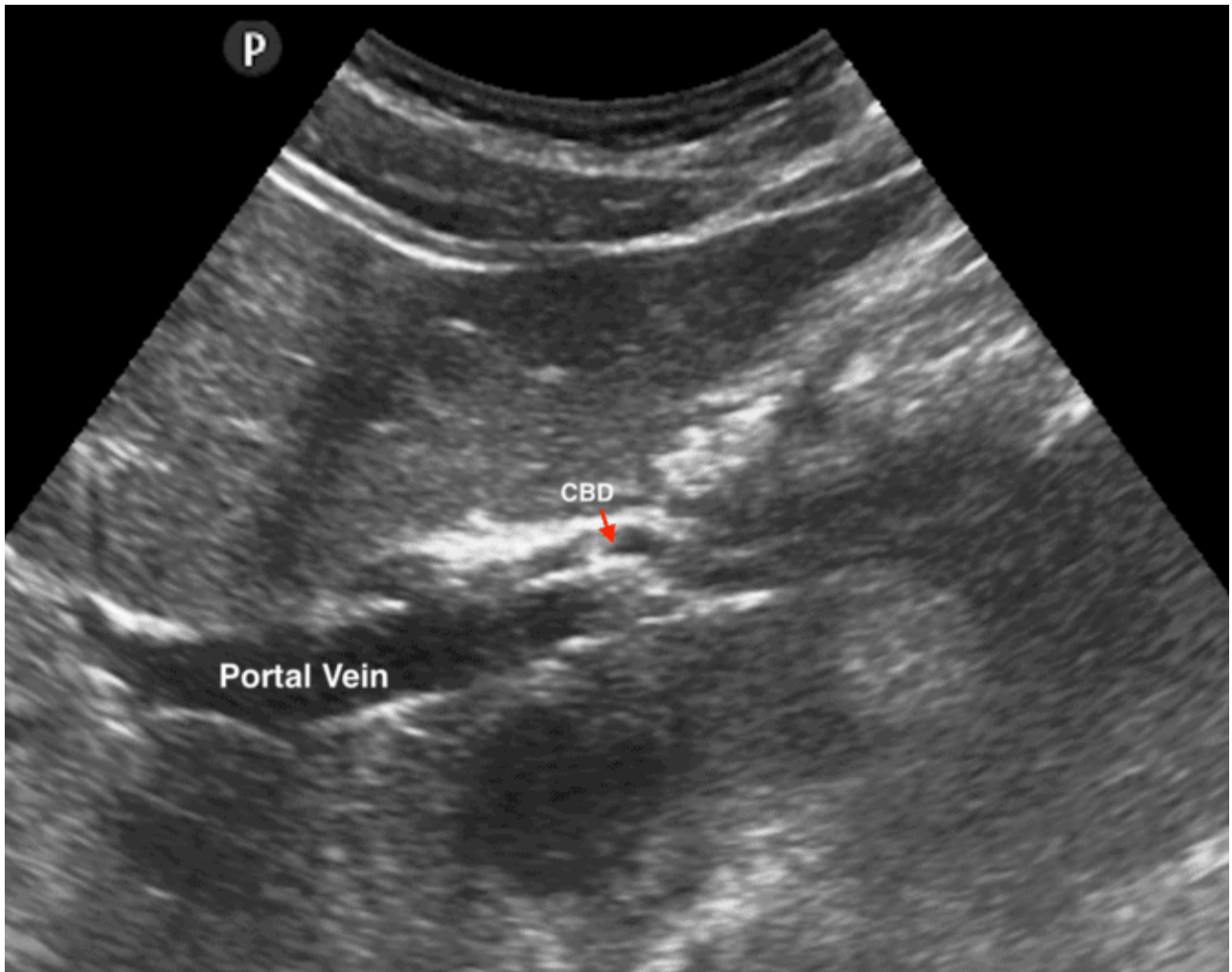
decubitus position as this will bring the gallbladder more into the midline for your to scan.

- **Position your probe in the epigastric midline** with the **indicator facing the patient's head**.
- In the midline, you will likely see the Main Portal vein in the long-axis, proximal to when it curves into the liver.



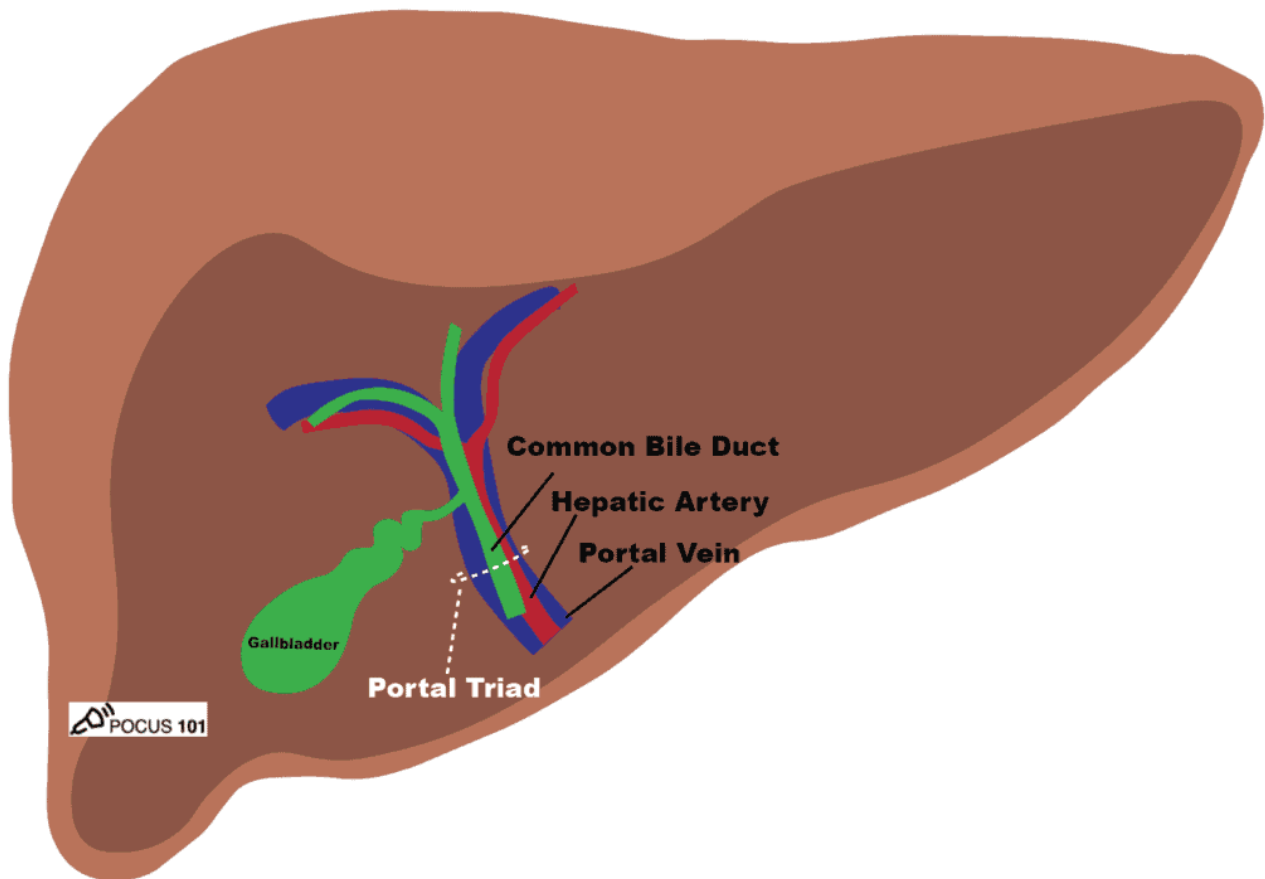


*Probe position to begin the gallbladder US exam.*



*The portal vein seen in long-axis. In this picture, the common bile duct is also seen adjacent to the portal vein, in the long-axis as well.*

**Slide laterally** (slowly) along the costal margin, toward the patient's right, until you can clearly see the **Portal Triad** (Portal vein, Hepatic Artery, and Common Bile Duct) in short axis.



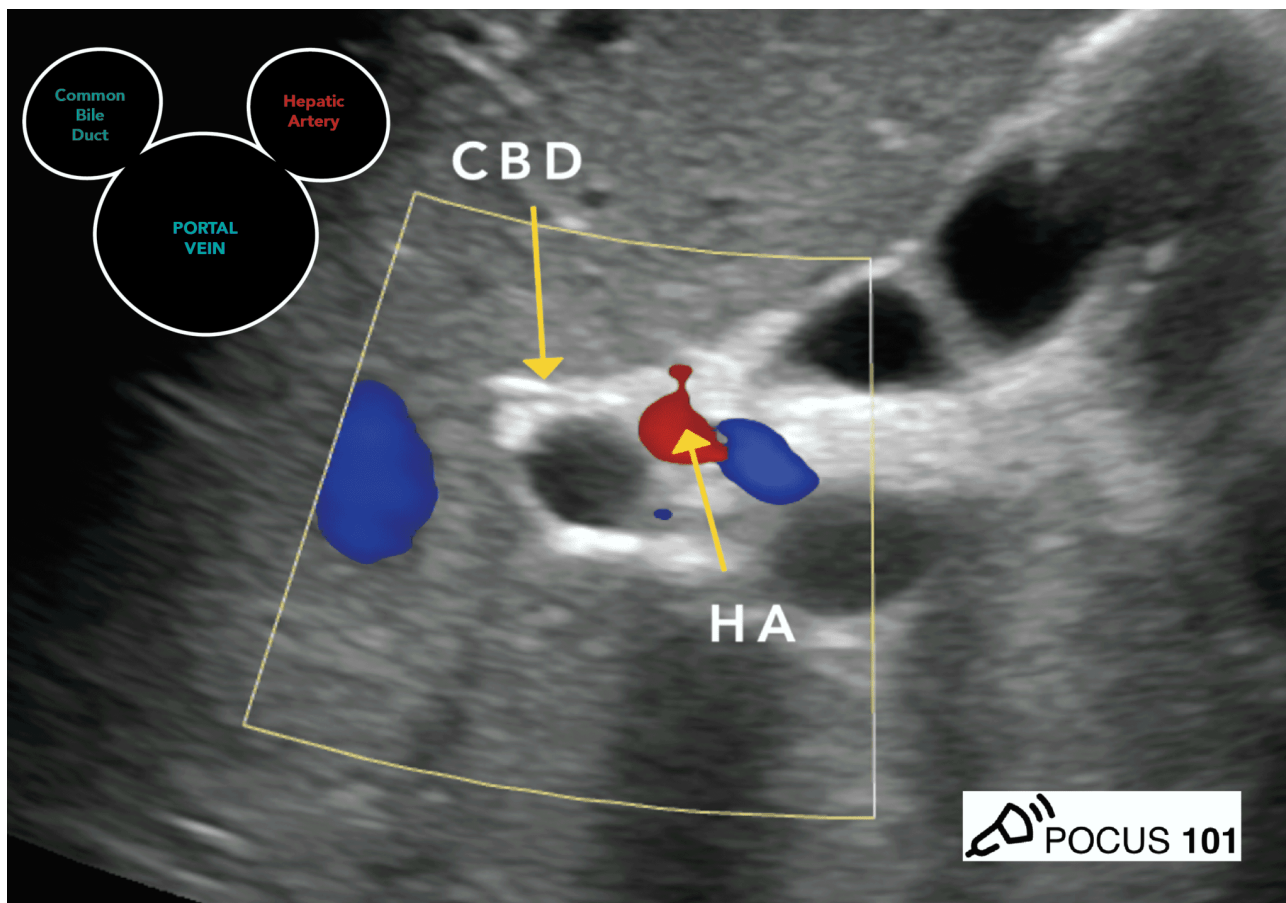
Portal Triad

The **Portal Triad** seen in the short-axis is known as the “**Mickey Mouse Sign**,” where the Common Bile Duct (CBD) and Hepatic Artery are the “ears,” and the portal vein is the “head.”



*Slide laterally to see the portal vein in the short axis.*





*The portal triad in the short axis is known as the “Mickey Mouse Sign.” The “ears” are the hepatic artery (HA) and common bile duct (CBD), and the “head” is the portal vein.*

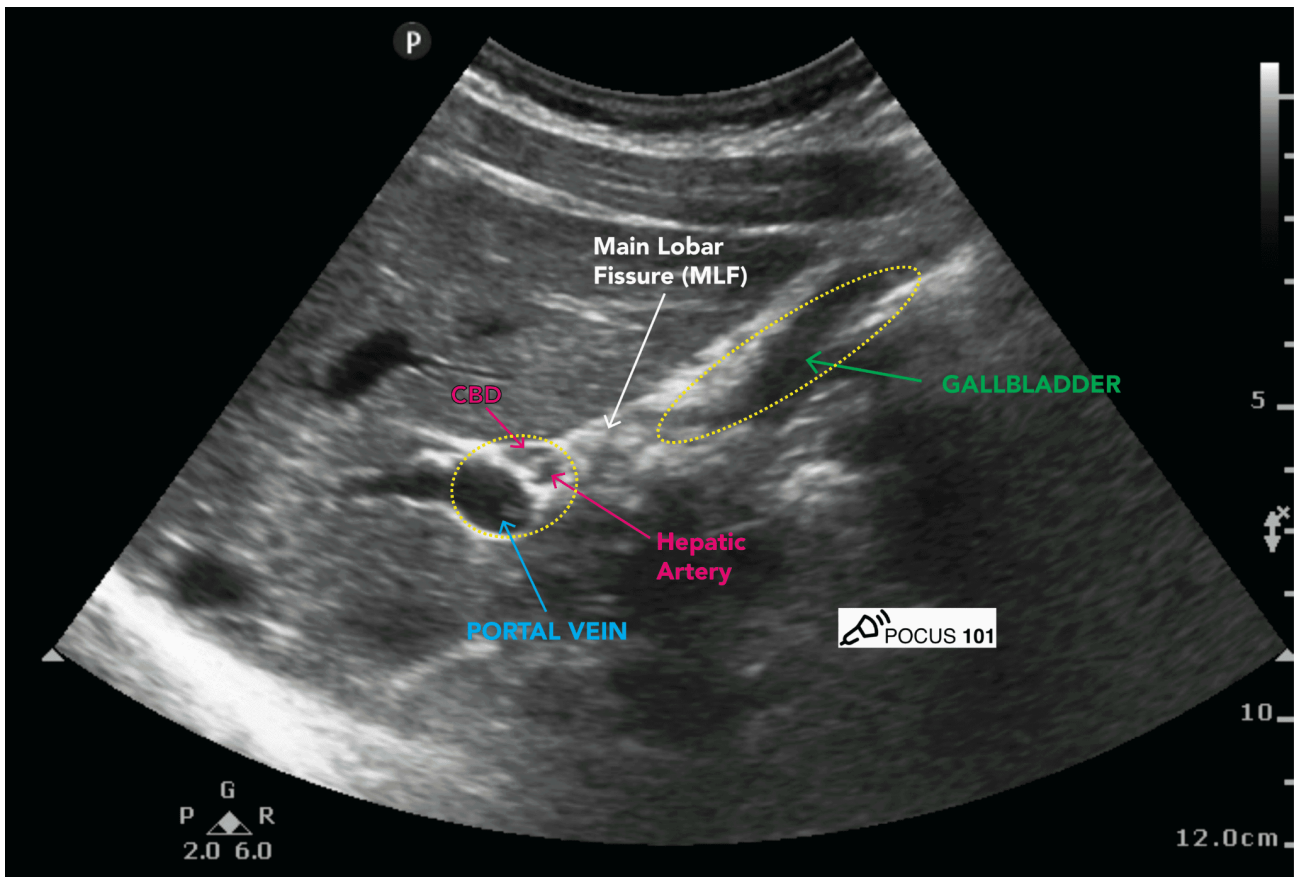
### **Visualize the gallbladder and main lobar fissure (MLF).**

- The gallbladder neck is attached to the **Main Lobar Fissure (MLF)**. Search adjacent to the MLF, on the right of the screen, by rotating your probe clockwise as shown below.
- This view is known as the “**Exclamation Point Sign**,” with the portal triad as the “point” and the gallbladder as the “line” of the exclamation.





*Rotating clockwise to find the MLF.*



*The gallbladder is seen attached to the MLF, adjacent to the portal triad. The “Exclamation Point Sign” is outlined by the yellow dots.*

### **Evaluate for gallstones.**

Tilt/fan your probe superiorly and inferiorly as shown in the GIF to search for cholelithiasis.

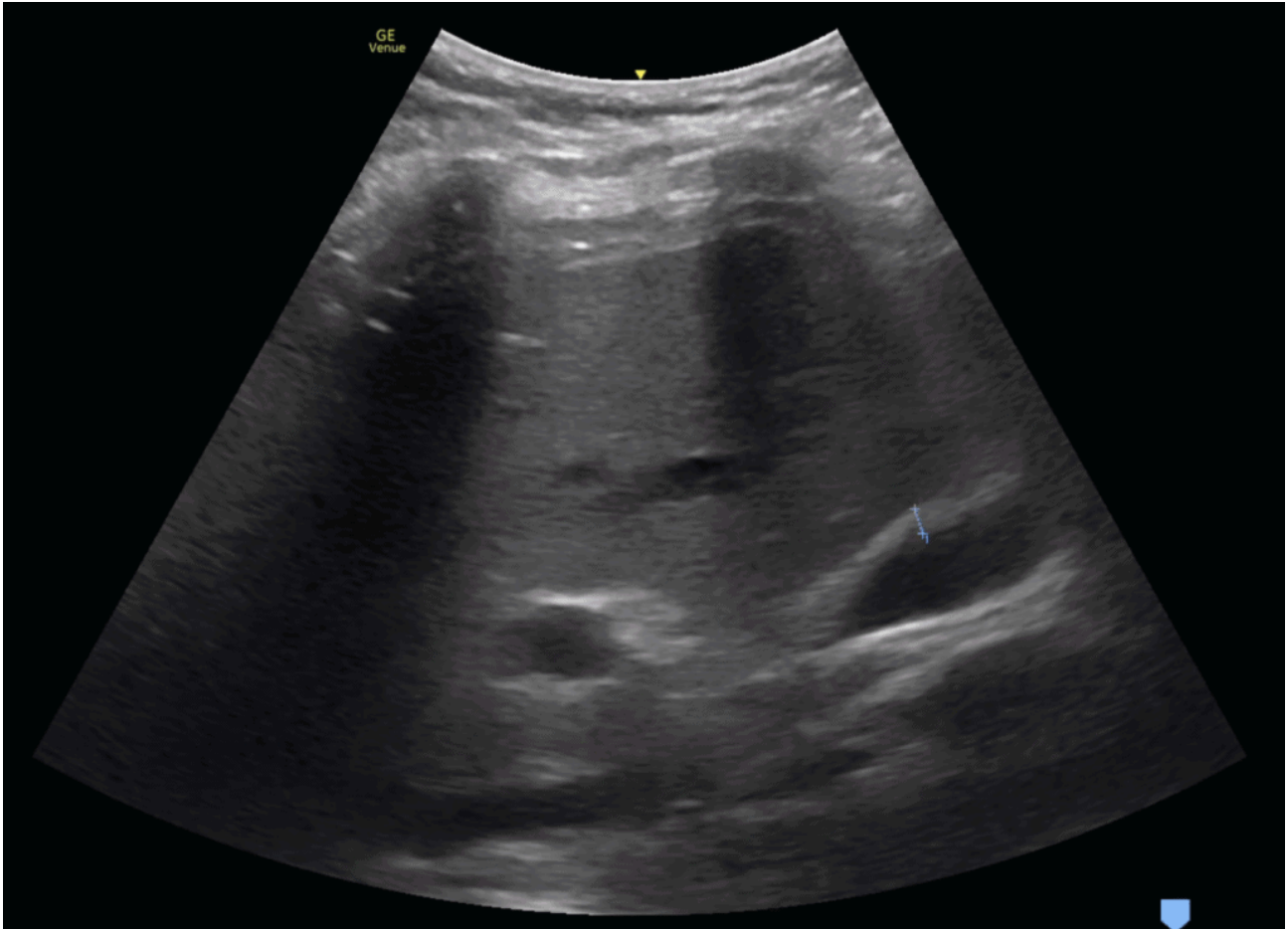




*Tilting/Fanning superiorly and inferiorly.*

**Measure the anterior gallbladder wall thickness.**

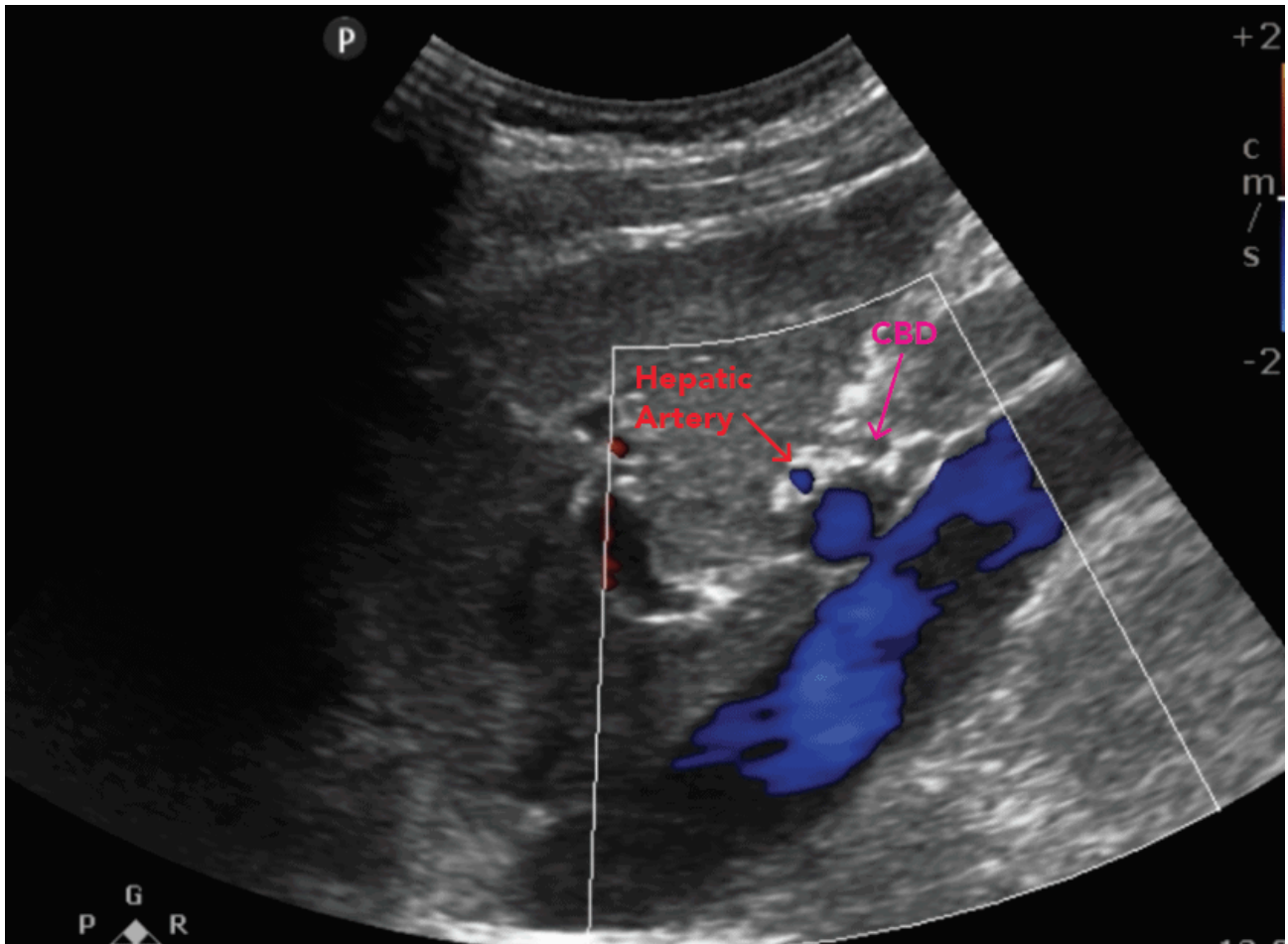
- Orient your calipers as pictured and measure the **anterior gallbladder wall thickness**.
- The normal anterior gallbladder wall thickness is **<3mm**
- *Editor's note: some resources use <5mm thickness to increase the specificity. Remember to use your hospital's guidelines when measuring.*



*Measuring the anterior gallbladder wall thickness.*

### **Evaluate the Common Bile Duct.**

- Return to the short-axis view of the portal triad (“Mickey Mouse Sign”) described above.
- **Use color Doppler to distinguish the common bile duct from the hepatic artery (proper).** Flow will be seen in the hepatic artery, while the common bile duct will remain anechoic with no pulsatile flow.

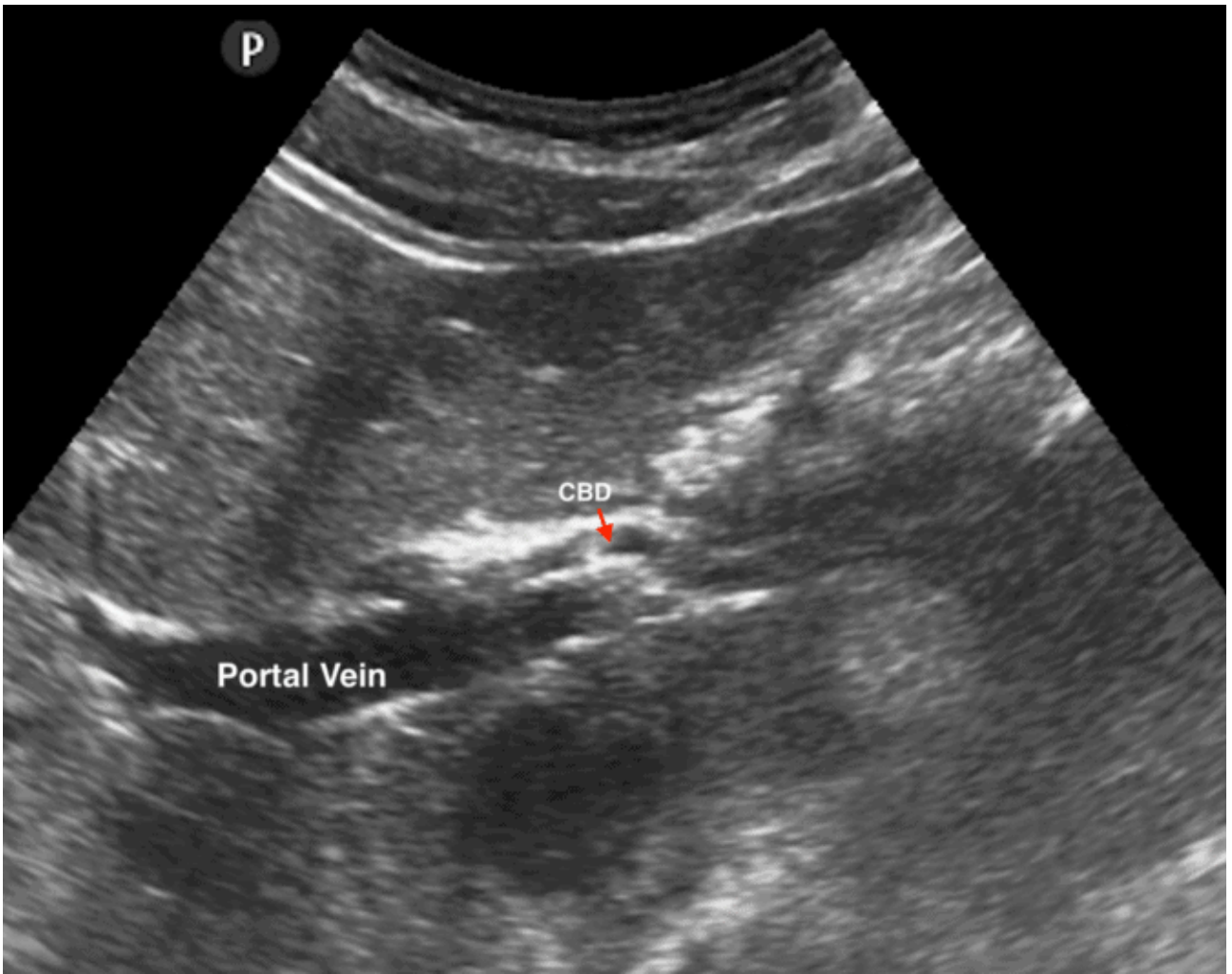


Flow seen in the IVC, portal vein (in short axis), and right hepatic artery – but not in the common bile duct, which will show no flow.

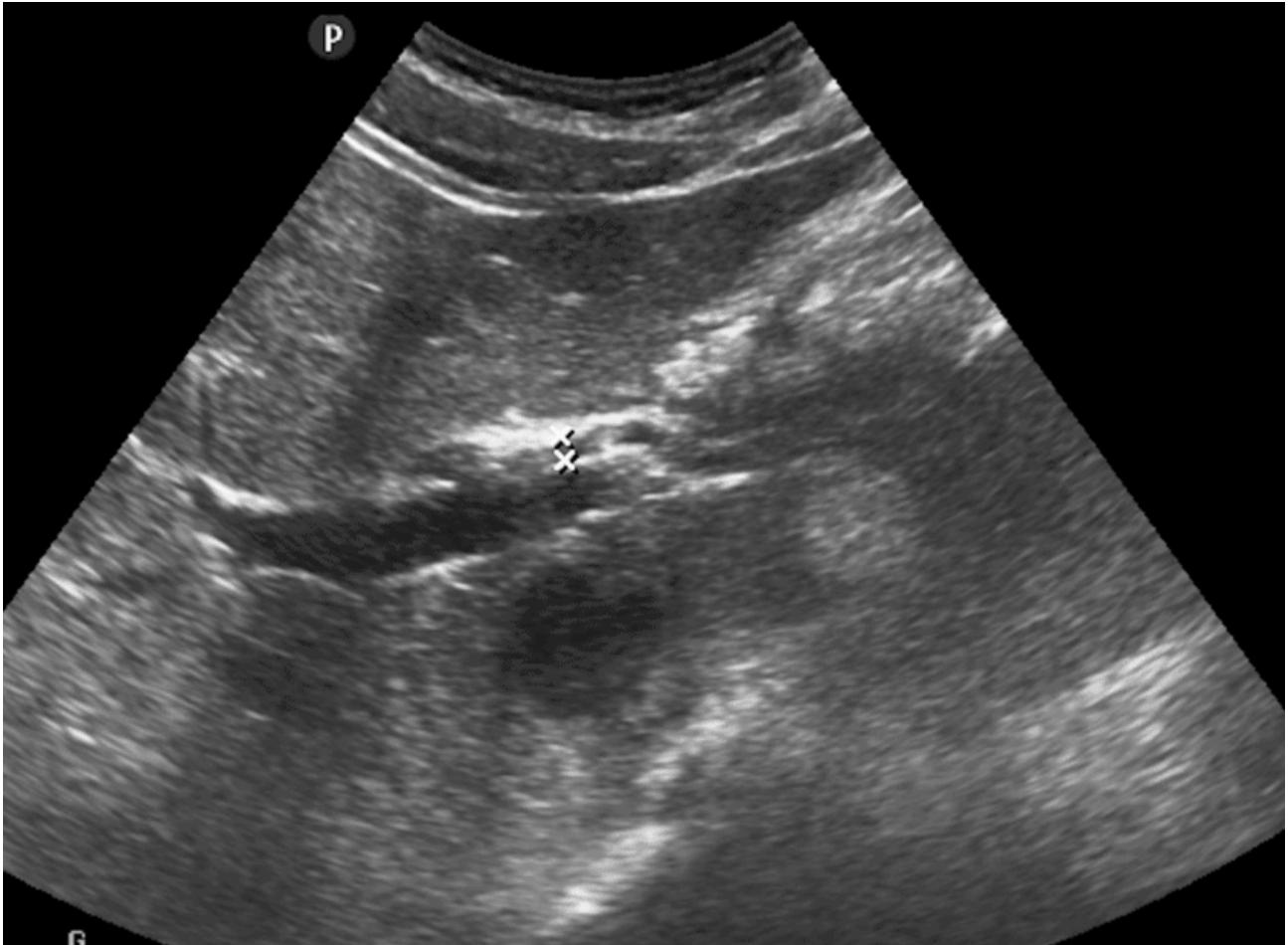
**Visualize the common bile duct in the long axis and measure the anteroposterior diameter.**

- From the short-axis view, rotate your probe 90° counter-clockwise
- Once you have identified the common bile duct in the long axis, **measure** it from **inner wall to inner wall** as shown below.
- **A normal CBD measurement is <5mm in the average population**
- However, with every decade of life after 50, the expected diameter increases by one millimeter. For example, a 60-year-old patient would be expected to have a CBD around 6mm, and a 70-year-old patient would be expected to have a CBD of 7mm.
- Also, patients that have had their gallbladder removed (cholecystectomy) can have normal CBD measurements of <10mm.





Portal Vein and Common Bile Duct



Common Bile Duct Measurement (inner to inner)

*POCUS 101 Tip: it can be difficult to visualize, let alone measure the CBD. Our quick tip is that if the CBD doesn't appear wider than 50% the portal vein diameter, it is unlikely any bile duct pathology is present.*

## Gallbladder/CBD Pathology

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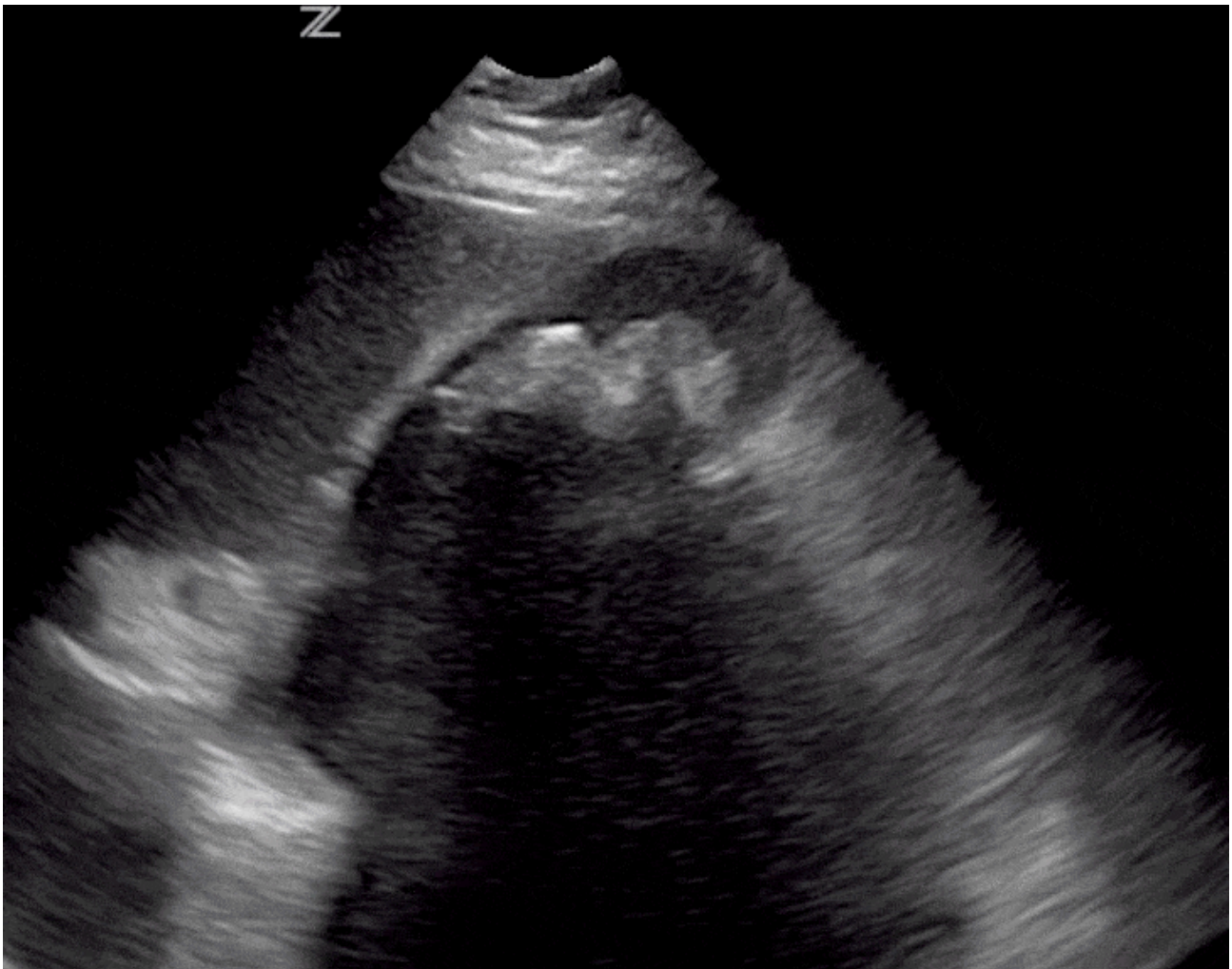
Here we'll cover the most common patient presentations and ultrasound findings for cholelithiasis, cholecystitis, and choledocholithiasis. Ultrasound is by far the preferred initial imaging study for assessing RUQ pain, and boasts high sensitivities and specificities for cholelithiasis and cholecystitis (Revzin).

### Cholelithiasis

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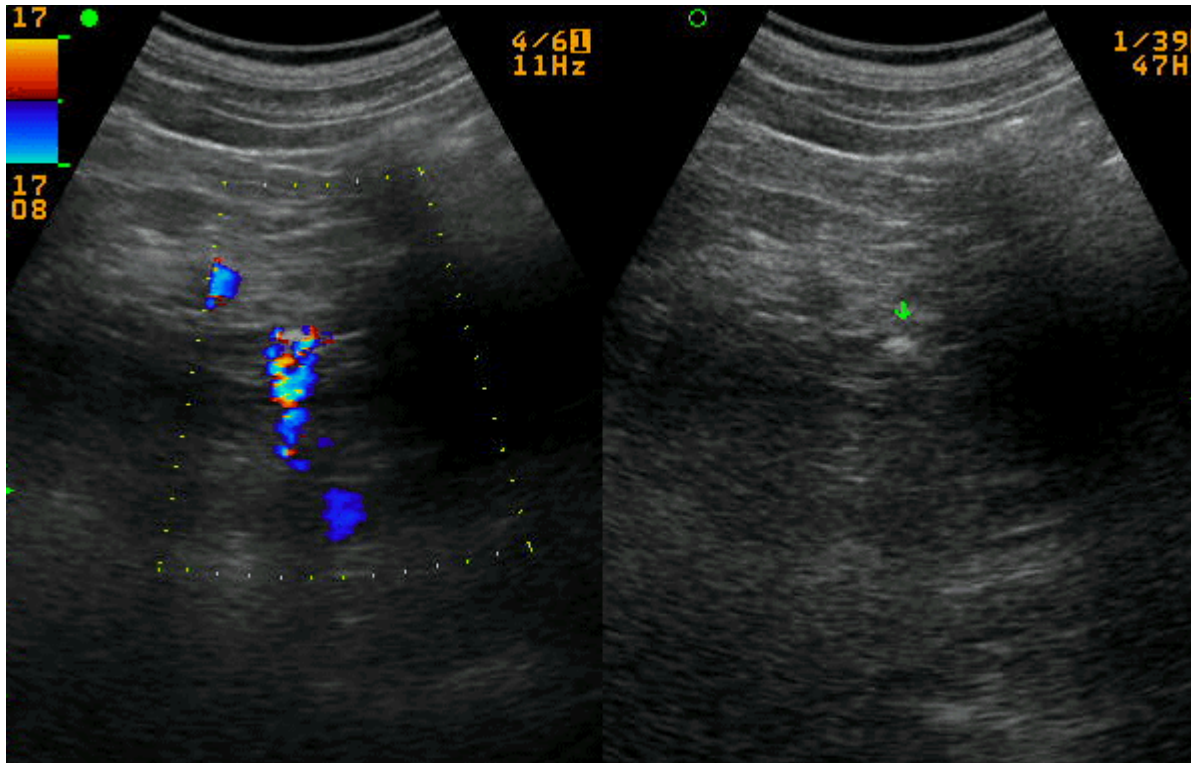
Cholelithiasis refers to gallstone formation anywhere along the biliary tree. Gallstones are present in up to 10% of the population, and only about 1/4 of patients are symptomatic. When symptomatic, the most common finding is biliary colic (post-prandial RUQ pain, especially following a fat-rich meal). Pain may radiate to tip of the right scapula, a finding known as Collin's sign. Nausea, bloating, belching, heartburn, and flatulence are also common (Bell).

Ultrasound is considered the gold standard for the detection of gallstones (Kothari), which will appear as highly **hyperechoic collections with posterior shadowing** within the gallbladder lumen. The stones may be gravity-dependent (“rolling stones”) and move within the lumen when the patient changes position.



*Large gallbladder calculi with posterior shadowing.*

Smaller stones may not be directly visible within the lumen, but may exhibit a “twinkling artifact” with color Doppler, pictured below. On the right, the tiny stone is visible in grayscale (green arrow).



Case courtesy of Dr Ayush Goel, [Radiopaedia.org](https://radiopaedia.org). From the case [rID: 29793](#)

## Cholecystitis on Ultrasound

Cholecystitis, or gallbladder inflammation, occurs when the gallbladder's drainage ducts are obstructed and the gallbladder becomes inflamed/infected. Cholelithiasis is the most common cause and primary risk factor for developing cholecystitis, implicated in up to 95% of cases of cholecystitis (Liau). Other causes (acalculous cholecystitis) include sludge (fine bile crystals mixed with proteinaceous debris and fluids) or tumors.

Patients with cholecystitis most commonly present with acute RUQ pain, persistent over  $\geq 6$  hours, that may radiate to the right shoulder. Nausea, vomiting, and fever are also common. Lab abnormalities can include leukocytosis and increased liver function tests.

The cardinal physical exam finding for cholecystitis is the traditional **Murphy's sign** on physical exam. With your fingers clasped under the right costal margin, a positive Murphy sign refers to a "catch" in the patient's breath when asked to inhale deeply, indicating pain when the gallbladder touches the examiner's fingers.

A **Sonographic Murphy sign**, however, refers to the presence of maximal tenderness when ultrasound probe pressure is applied over the visualized gallbladder on ultrasound. This is a very sensitive finding for cholecystitis (Bree).

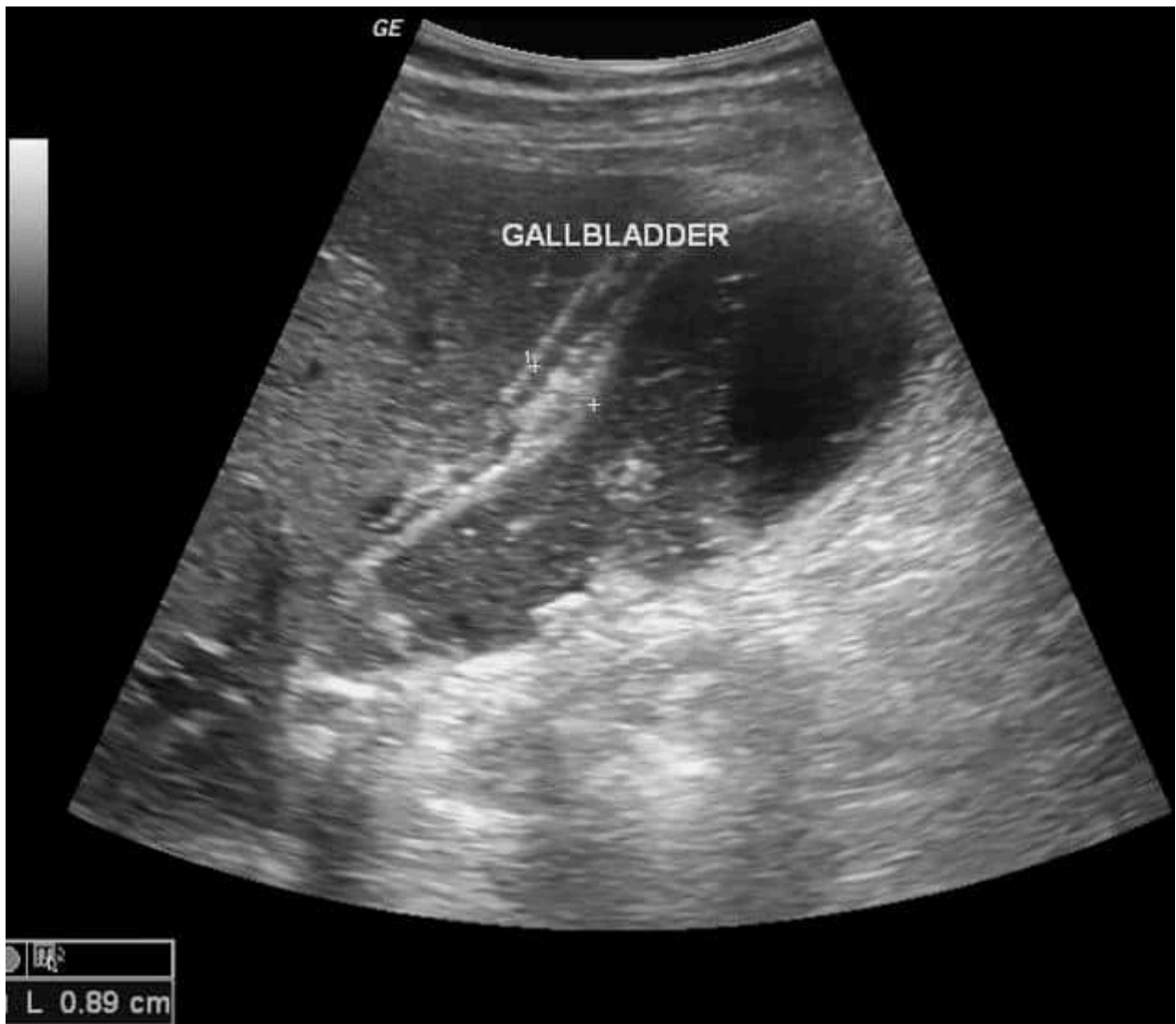




*Technique for eliciting the Sonographic Murphy Sign.*

Important secondary findings include **cholelithiasis (gallstones)**, **anterior gallbladder wall thickening >3mm**, and **pericholecystic fluid**.

Other less specific findings are **gallbladder sludge** and **gallbladder distension** (>4cm transverse and >9cm longitudinal).



*Distended gallbladder with calculi, sludge, a thickened anterior wall, and pericholecystic collection.*

Case courtesy of Dr Maulik S Patel, [Radiopaedia.org](http://Radiopaedia.org). From the case [rID: 20542](https://radiopaedia.org/cases/20542)

## Choledocholithiasis

Choledocholithiasis occurs when gallstones lodge in the bile ducts and bile can no longer flow, instead backing up into the liver. Most cases are symptomatic and require removal of the stones since ascending cholangitis (acute bacterial infection of the biliary tree) is a common complication with high morbidity/mortality.

Patients with choledocolithiasis usually present with biliary colic and obstructive jaundice, ascending cholangitis, and potentially acute pancreatitis (Hartung).

On ultrasound, look out for calculi in the gallbladder and/or CBD, and a **dilated common bile duct**. Any CBD **wider than 5mm** should raise suspicion, especially when paired with a suggestive history. However, with every decade of life past 50, the expected diameter increases by one millimeter. For example, a 60-year-old patient would be expected to have a CBD around 6mm, and a 70-year-old patient would be expected to have a CBD of 7mm.



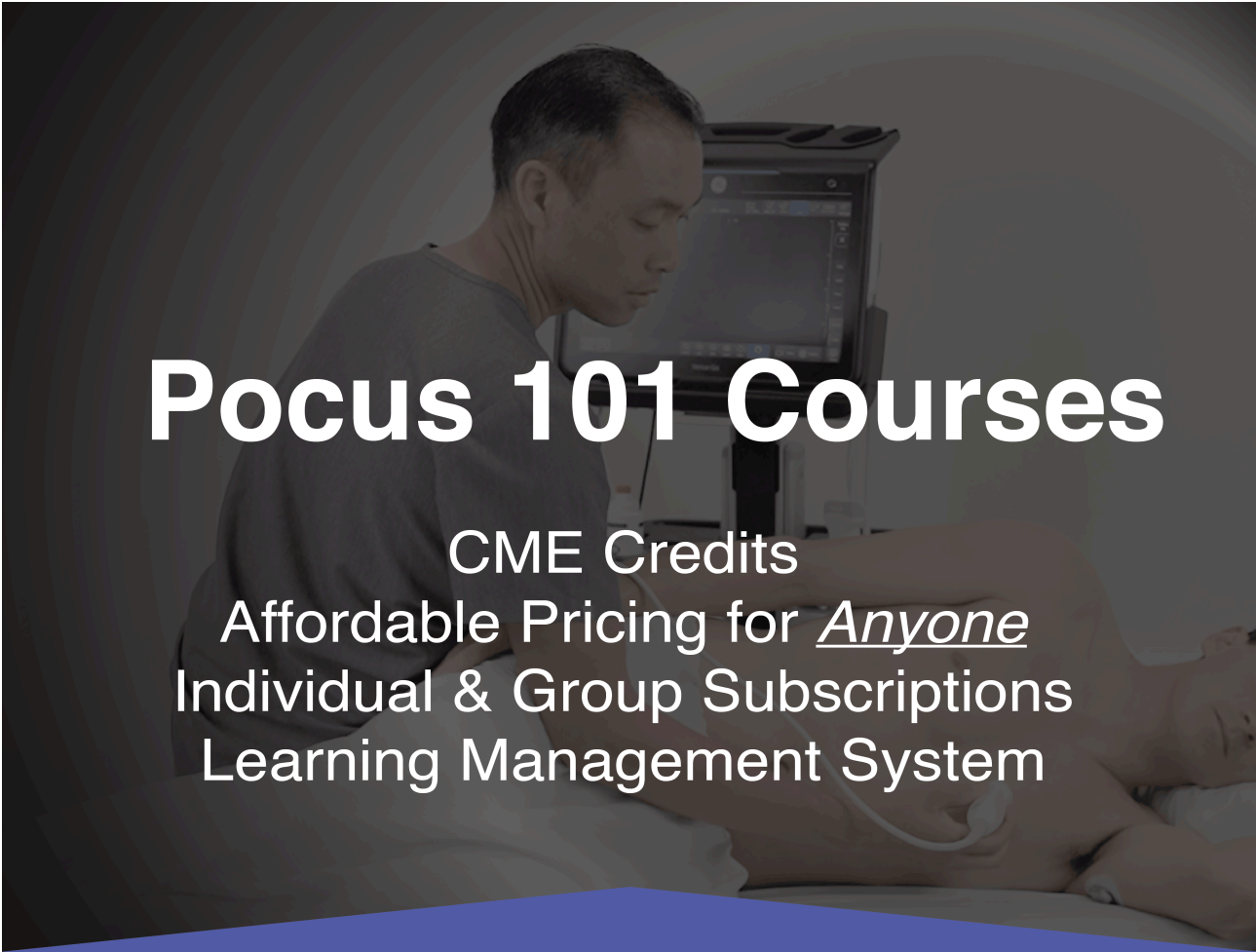
*Large calculi in the gallbladder (top) and CBD (bottom).*

Case courtesy of Dr David Cuete, [Radiopaedia.org](http://Radiopaedia.org). From the case [rID: 43915](https://radiopaedia.org/cases/43915)

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## Spleen Ultrasound

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The spleen is imaged on the patient's left side using ultrasonography. As with the liver, the spleen can be used as an acoustic window to assess other important structures, such as the aorta and IVC. Here, you'll learn how to visualize the spleen and measure its craniocaudal length to evaluate for splenomegaly.

## Spleen Ultrasound Protocol

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- Place your probe in the **7th-8th intercostal space** on the patient's left side, with the **indicator facing the patient's head**.
- The spleen lies quite posteriorly, so you'll want to be "**knuckles-to-bed**" to properly visualize it.





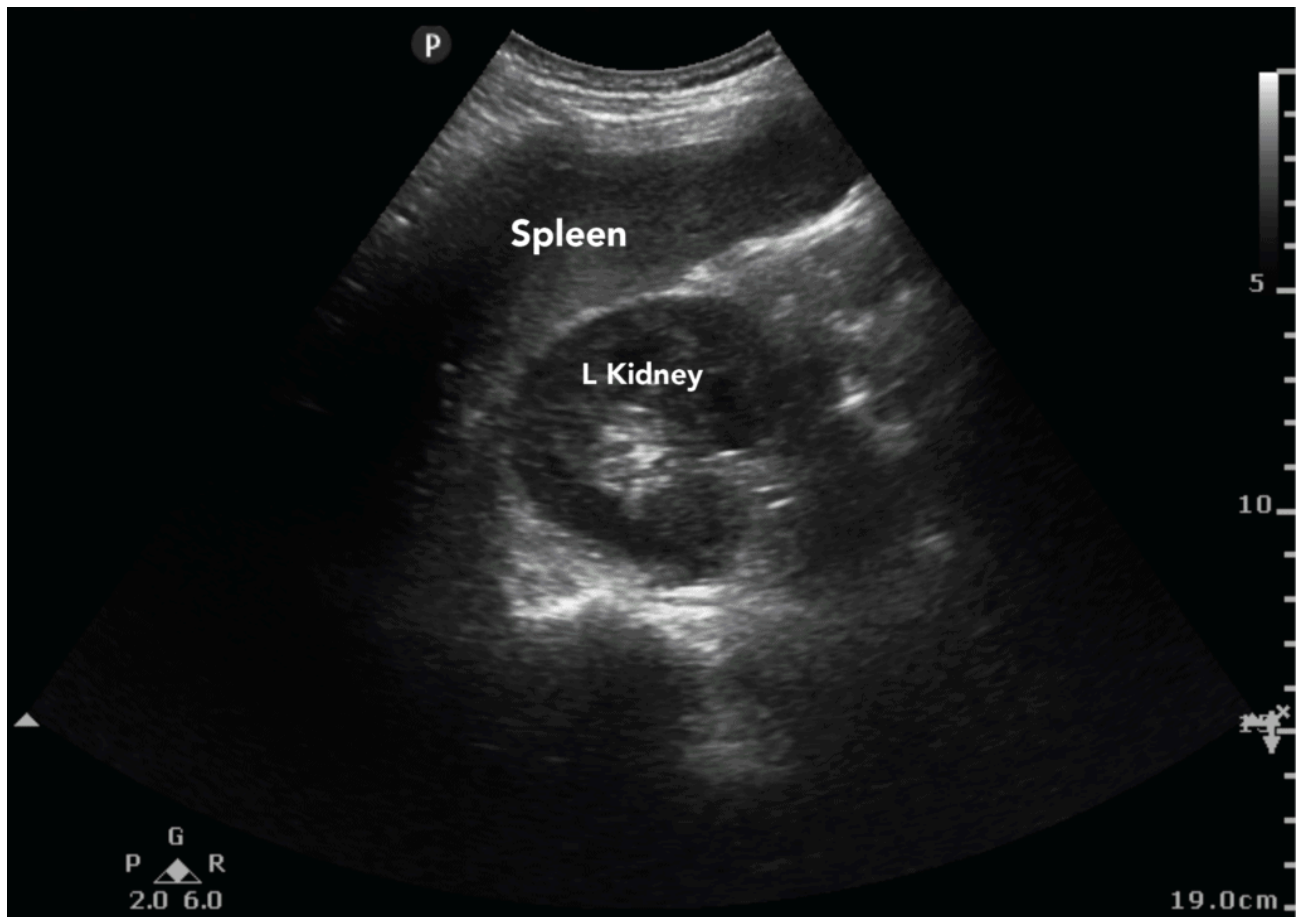
Spleen Ultrasound Probe Positioning. Notice “knuckles to the bed”

**Visualize the spleen and adjacent left kidney.**

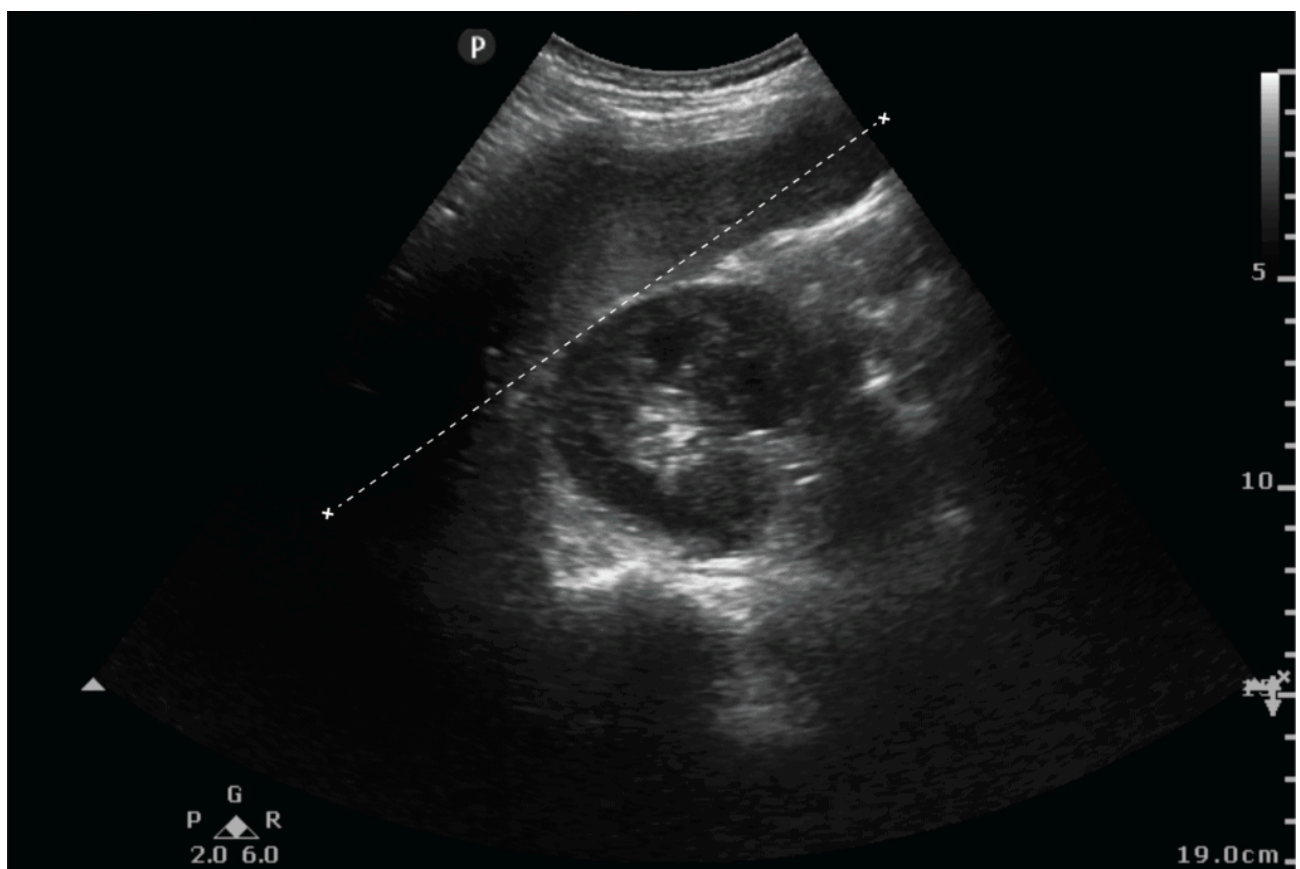
If the ribs are casting a shadow over your window, try having the patient hold a deep breath to further widen the intercostal spaces. Align your probe obliquely with the intercostal space by rotating slightly clockwise.

**Measure the spleen in the craniocaudal dimension.** This is to evaluate for splenomegaly.

- If you cannot see the superior and inferior boundaries of the spleen, estimate as shown below.
- A normal spleen measurement is <12-14cm.



Ultrasound of Spleen



Spleen Measurement

Here is a short video going over how to evaluate the spleen using abdominal ultrasound:



[Watch on YouTube](#)

## Splenomegaly

There are numerous etiologies of splenomegaly, which can be broadly categorized as hematological (anemias, hematological neoplasms), hemodynamic (such as portal vein obstruction), infectious (especially viral), metabolic, neoplastic (non-hematological metastases), and owing to connective tissue disorders.

Since the etiologies are so varied, the associated manifestations will also vary widely from patient to patient. In rare cases, you may be able to appreciate a palpably enlarged spleen on physical exam. Patients may also complain vaguely of dull LUQ pain, fatigue (from anemia), and abdominal fullness.

A healthy spleen has a homogeneous echogenicity, smooth capsular contours, and a **craniocaudal length no more than 12cm in an adult** (although taller males may have a spleen up to 14cm long). Any larger, and your patient likely has splenomegaly.

Other findings in splenomegaly include a rounded inferior splenic surface – note above how the spleen sort of “hugs” the kidney and is convex inferiorly, whereas in the image below, the bottom has become concave with respect to the kidney, also dwarfing it in size.



*Severely enlarged spleen measuring 22cm, showing a more concave inferior surface, and massive dimension relative to the adjacent kidney.*

Case courtesy of Dr Ian Bickle, [Radiopaedia.org](http://Radiopaedia.org). From the case [rID: 52825](#)

## Appendix Ultrasound

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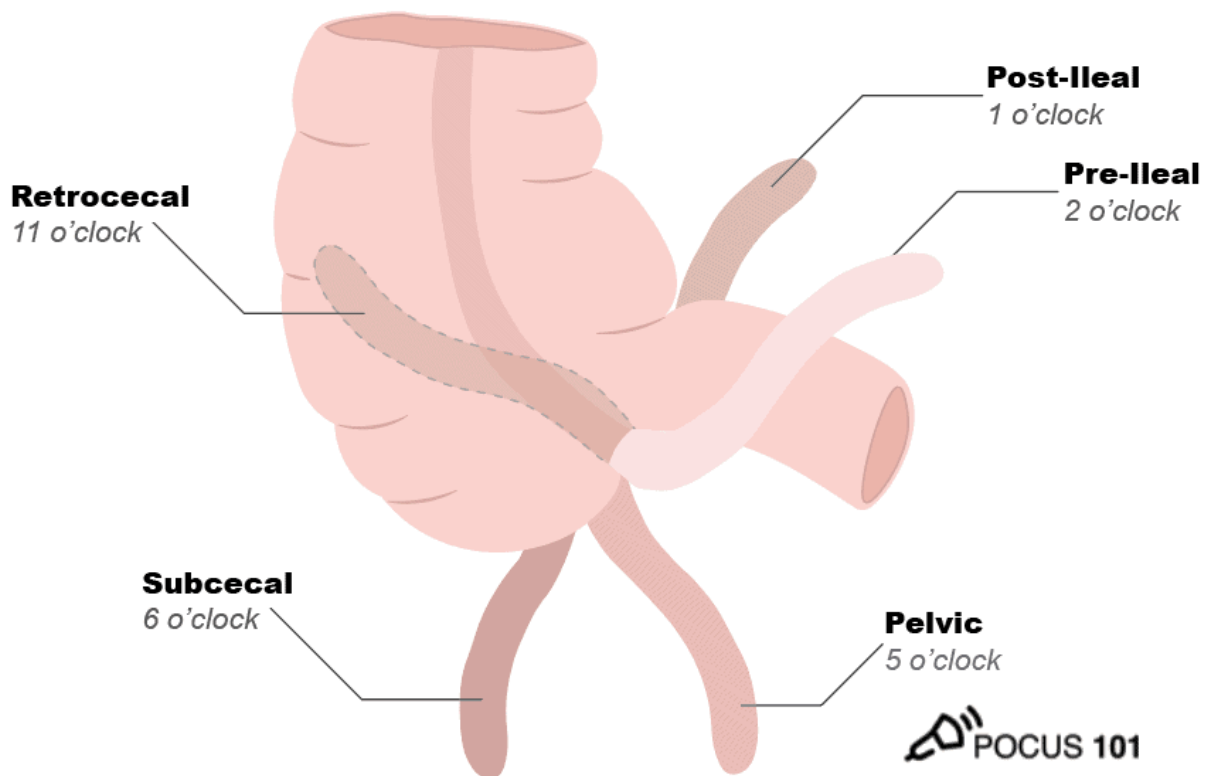
As one of the most common conditions prompting an ED visit, appendicitis is a pathology you are certain to encounter. And it's a pathology you certainly don't want to miss, given the complications associated with perforation.

Indications for performing an ultrasound exam of the appendix include the following common signs and symptoms:

- Focal RLQ pain
- Rebound tenderness/signs of peritoneal irritation
- Pelvic pain
- Fever
- Elevated WBC count

The appendix is located in the right lower quadrant, attached to the cecum. However, its location can vary widely in its anatomic position, as pictured below.





Appendix/Appendicitis locations

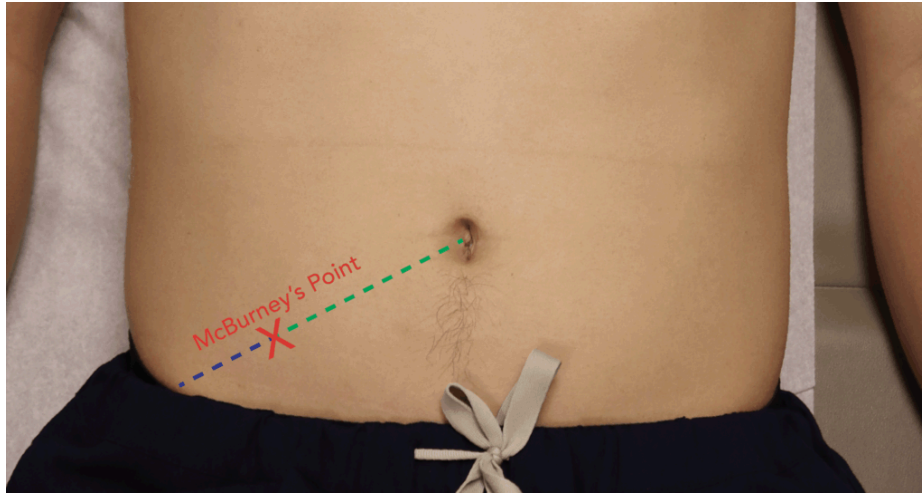
## Appendicitis Ultrasound Protocol

Acute appendicitis is the most common abdominal emergency, representing 250,000 cases annually, and a lifetime prevalence among men and women of 9% and 7%, respectively (Mostbeck).

However, diagnosing appendicitis is a challenge, despite its prevalence. Symptoms are often non-specific, overlapping with various abdominal pathologies. And finding the appendix can be tricky, especially since the appendix can be located retroceally.

This makes ultrasound less sensitive (~84%) for detecting appendicitis – failing to identify the appendix does not exclude it. However, POCUS is highly specific for appendicitis (~96%), and it should be your first-line imaging modality when you suspect it (Al-Khayal).

The most common presenting symptom of appendicitis is abdominal pain. The pain starts as dull periumbilical pain, then localized about a day later as sharp, stabbing pain at **McBurney's point**, or 1/3 the distance of a line drawn between the anterior superior iliac spine and the umbilicus.



McBurney's Point

There are two ways to go about the appendix POCUS exam, described below. We recommend starting with the “point of maximal tenderness” method, then using a more systematic approach to finding the appendix if the first method isn’t successful.

**First, position your patient and select your probe.**

- Probe selection: **linear probe** for a child/thin adult, or **curvilinear probe** for any patient with a large BMI.
- Preset: **Superficial/Abdomen**

### Option 1: Point of Maximal Tenderness

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This is our preferred method when you strongly suspect appendicitis based on your patient’s history and physical exam.

**Have your patient point, as precisely as possible, to “where it hurts” – the point of maximal tenderness.**

- Place the probe on the point of maximal tenderness with the indicator towards the patient’s right.
- Attempt to locate the appendix as a non-compressible structure with a blind-ended pouch.
- Rotate the probe to attempt to find the appendix in long and short-axis views.
- Look for the [signs of Appendicitis on Ultrasound described below](#).

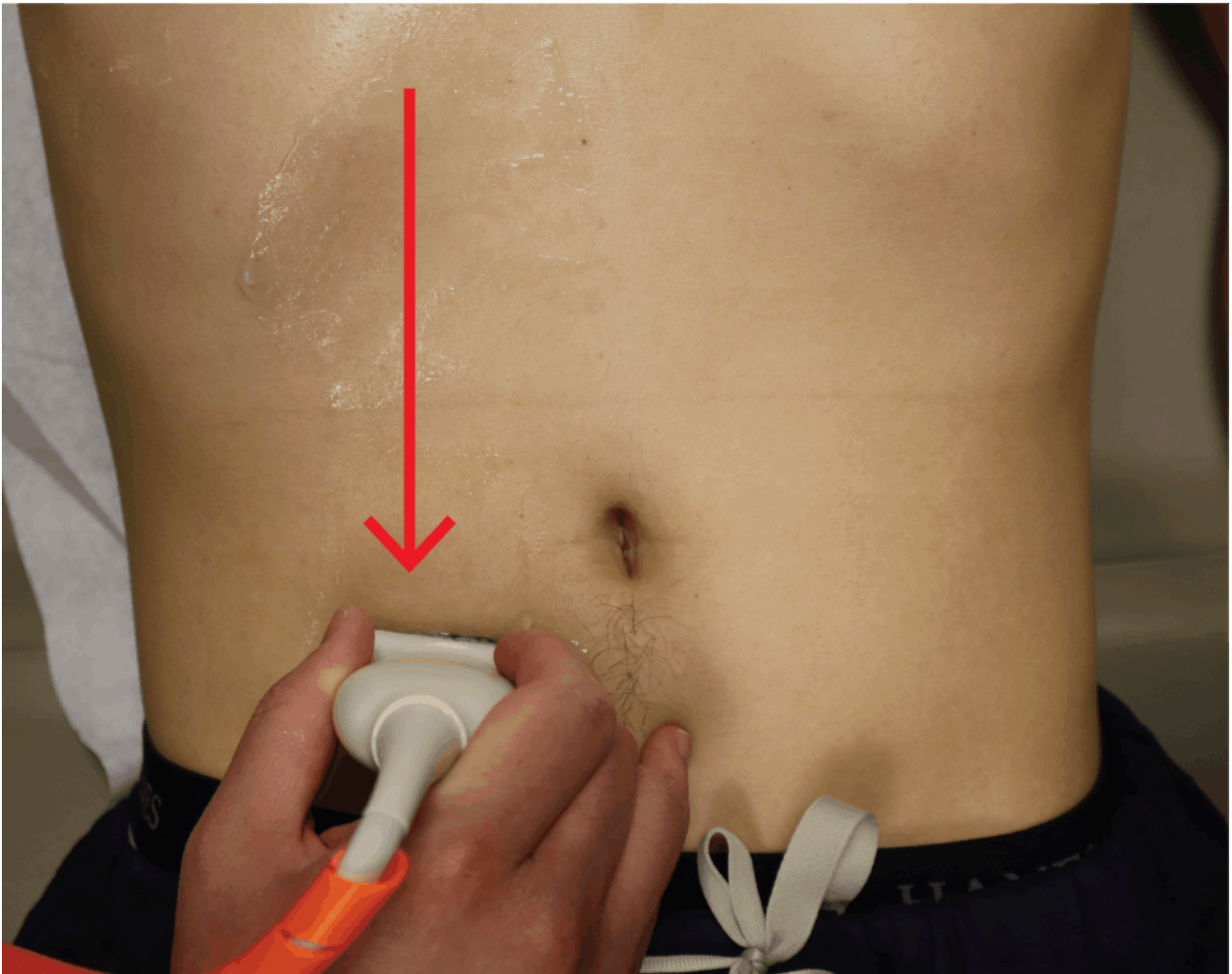
### Option 2: Systematic Approach

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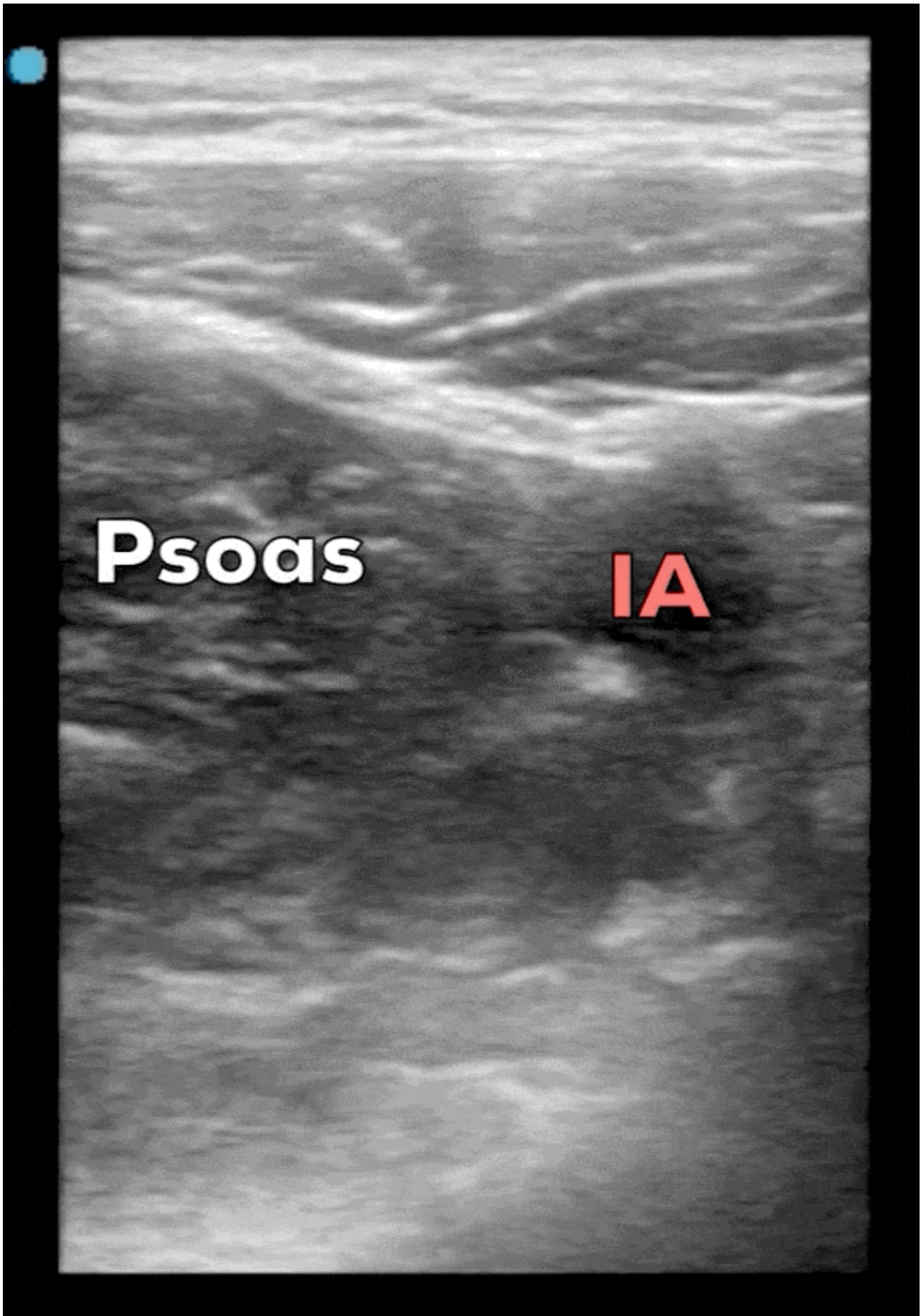
If you prefer a more organized approach, **try starting in the RUQ**, again with the probe in the transverse orientation with the **indicator facing the patient’s right**.

- Trace the ascending colon downward until reaching the cecum, and have identified the psoas major and the iliac vessels, as seen below.

- Now, search slightly medial to the psoas muscle for the [ultrasound findings of appendicitis described below](#).



Sliding down the Ascending Colon to look for the Appendix with Ultrasound



Landmarks for Appendicitis on Ultrasound – Psoas Muscle and Iliac Artery (IA)

*POCUS101 Tip: if bowel gas is obscuring the image, use graded compression, and watch the patient's eyes for any signs of discomfort as you apply pressure. The appendix can be*

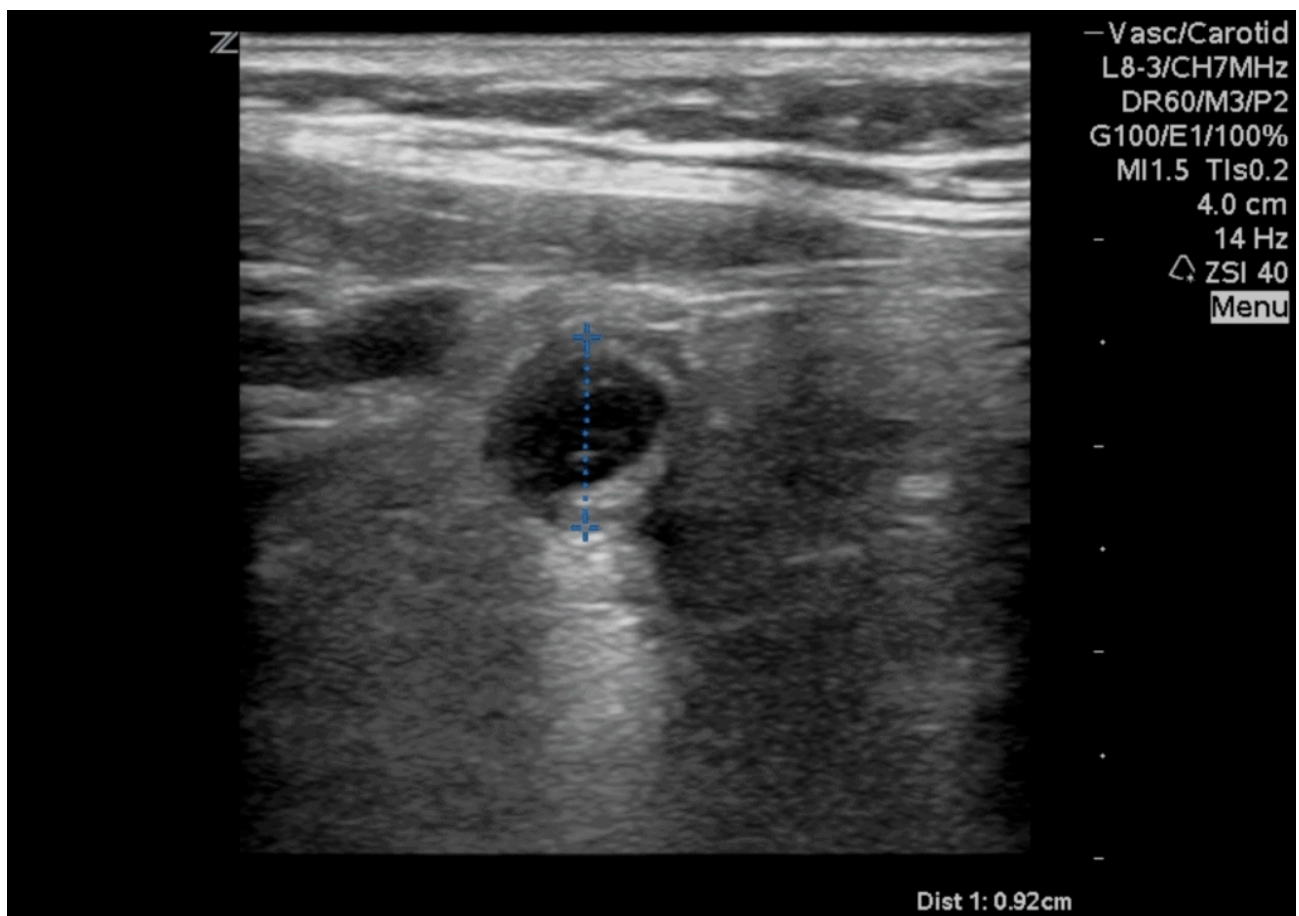


*especially difficult to visualize in patients with a large habitus. Keep in mind that the appendixes can be retrocecal, thus will be obscured from view when there is gas in the ascending colon.*

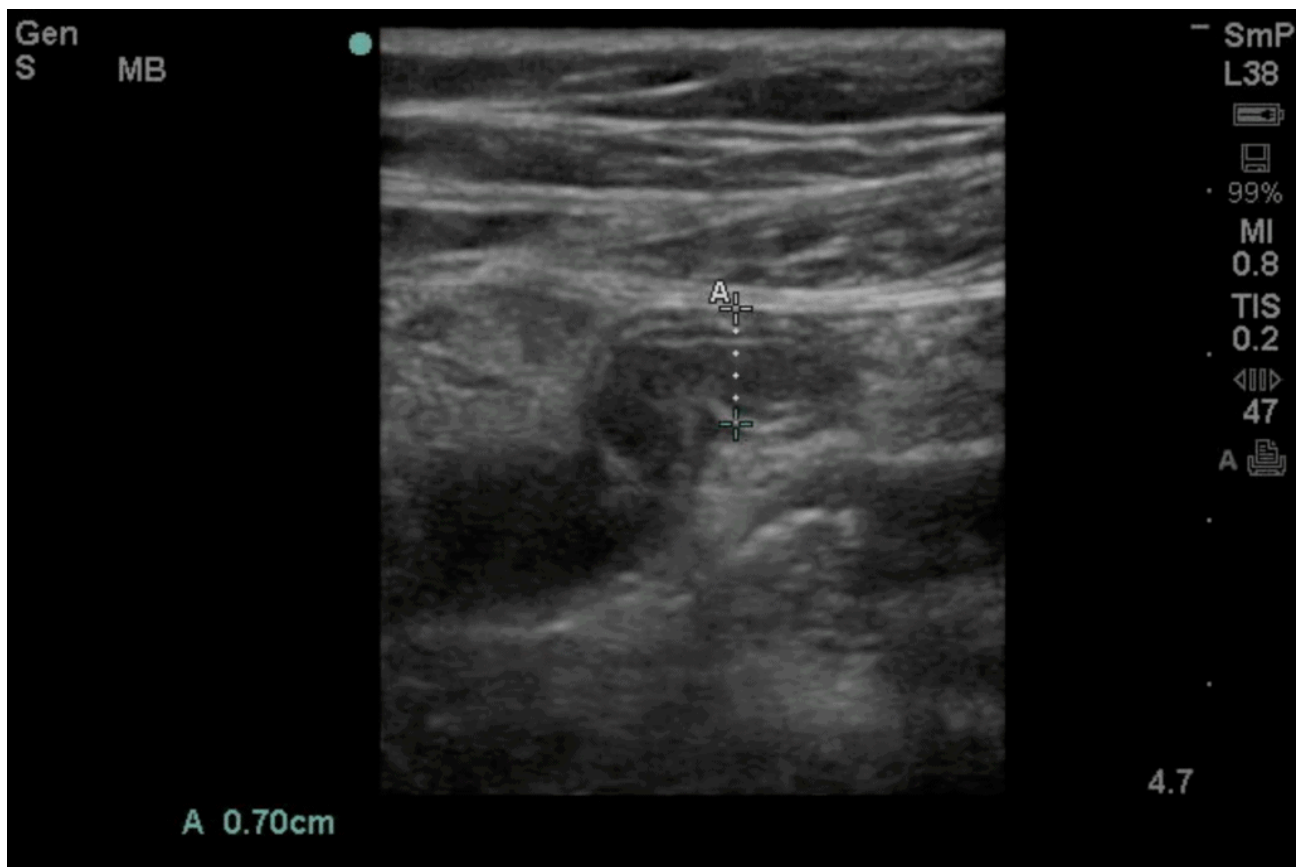
### Appendicitis Ultrasound Findings:

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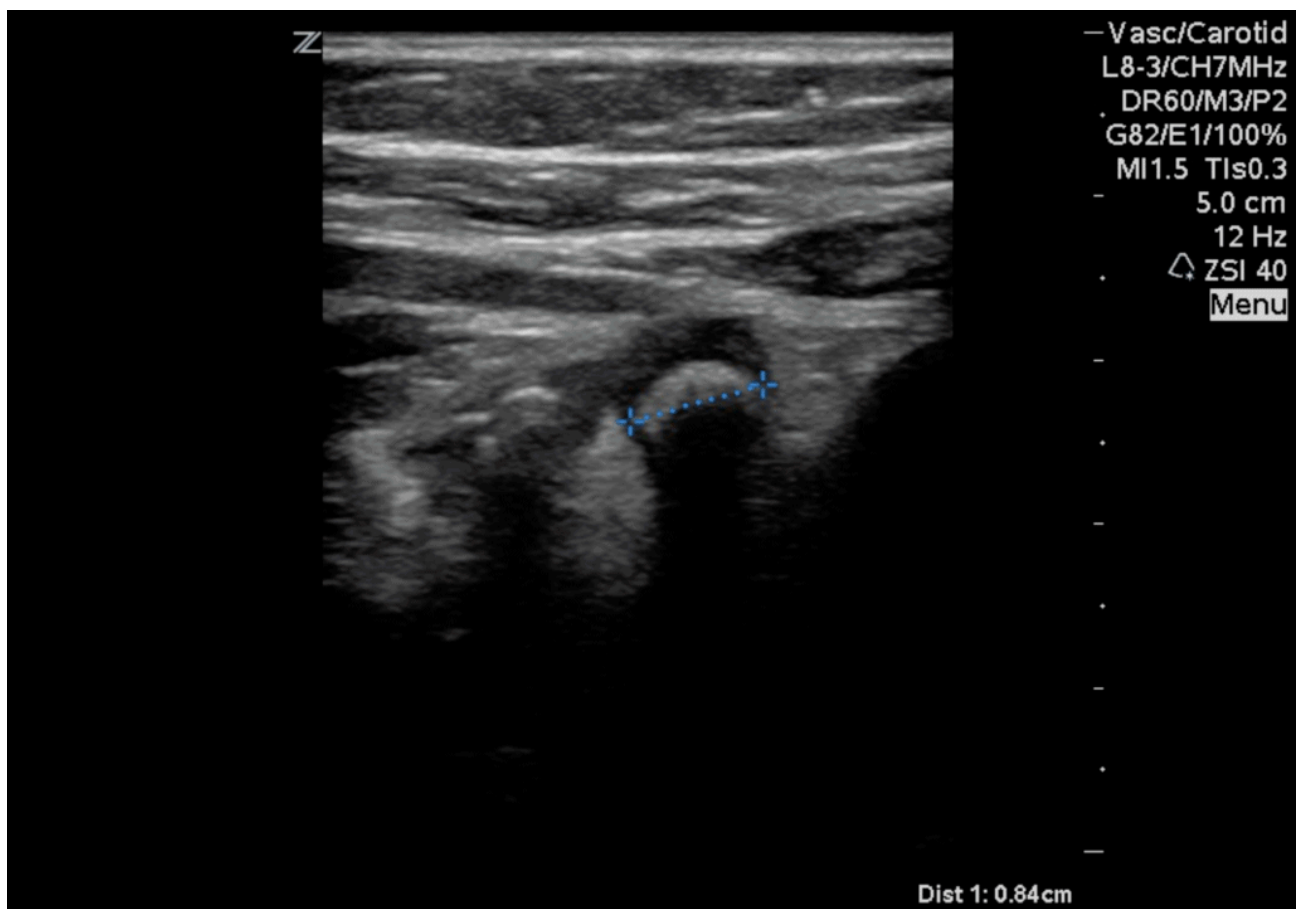
- Non-compressibility of the appendix (unless perforated)
- Blind ended pouch
- Diameter of the appendix > 6 mm
- Single wall thickness  $\geq 3$  mm
- Target sign
- Hyperechoic appendicolith with posterior shadowing



*Dilated appendix with a diameter >6mm and "Target Sign"*



*Dilated appendix with blind ended Pouch*



*Appendicolith in an appendix with a diameter > 6cm.*

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## Abdominal Free Fluid Ultrasound

Evaluating for free fluid is a common abdominal ultrasound application. We will go over the most common locations to detect free fluid using abdominal ultrasound.

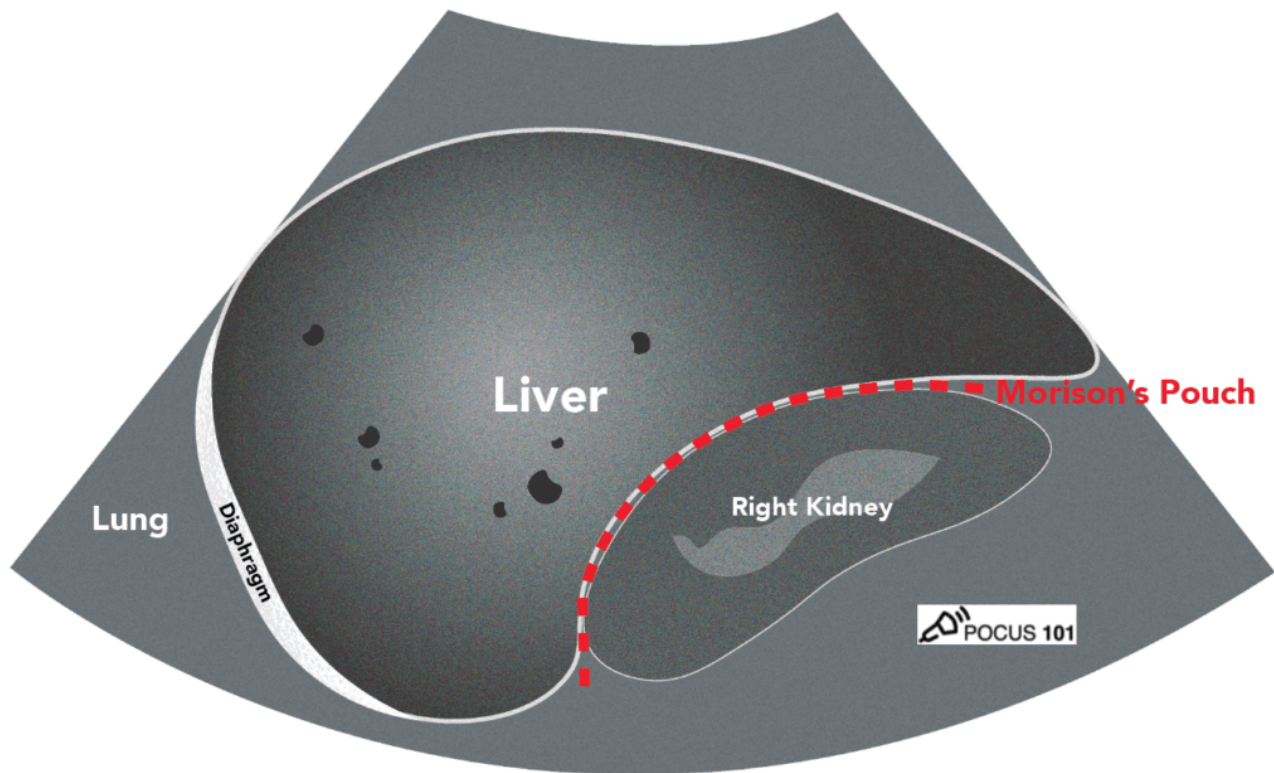
Using these techniques, you'll be able to diagnose ascites and hemoperitoneum in trauma patients ([read the complete eFAST exam tutorial HERE](#)). Keep in mind that blood will look more echogenic as hemoperitoneum progresses and begins to coagulate.



## Right Upper Quadrant (RUQ) – Morison's Pouch

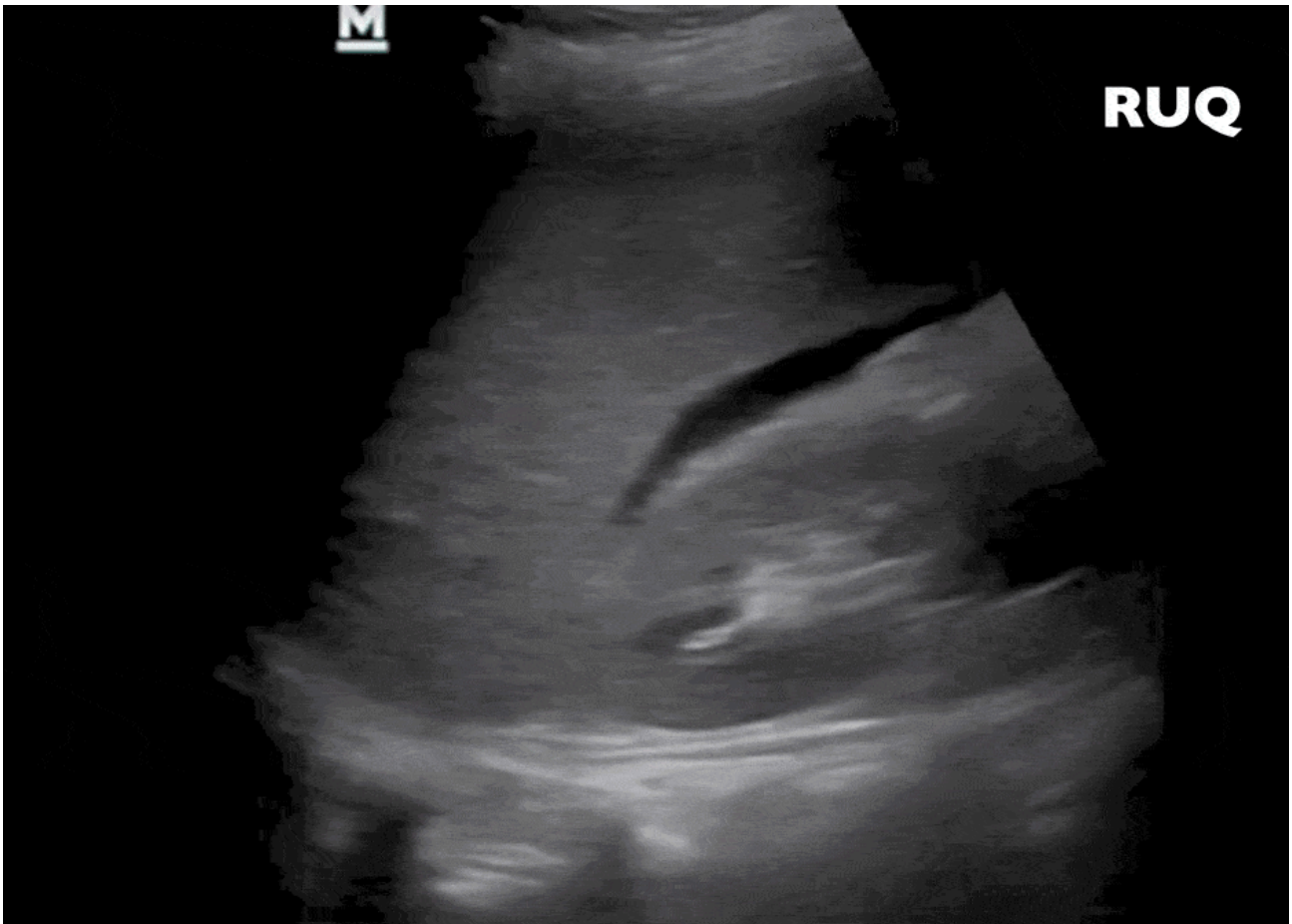
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Using the technique to scan the liver described above, you can detect free abdominal fluid in the **hepatorenal recess (Morison's Pouch)**. Normally, the recess is only a potential space, and no anechoic fluid should be seen between the liver and kidney.



Morison's Pouch Illustration



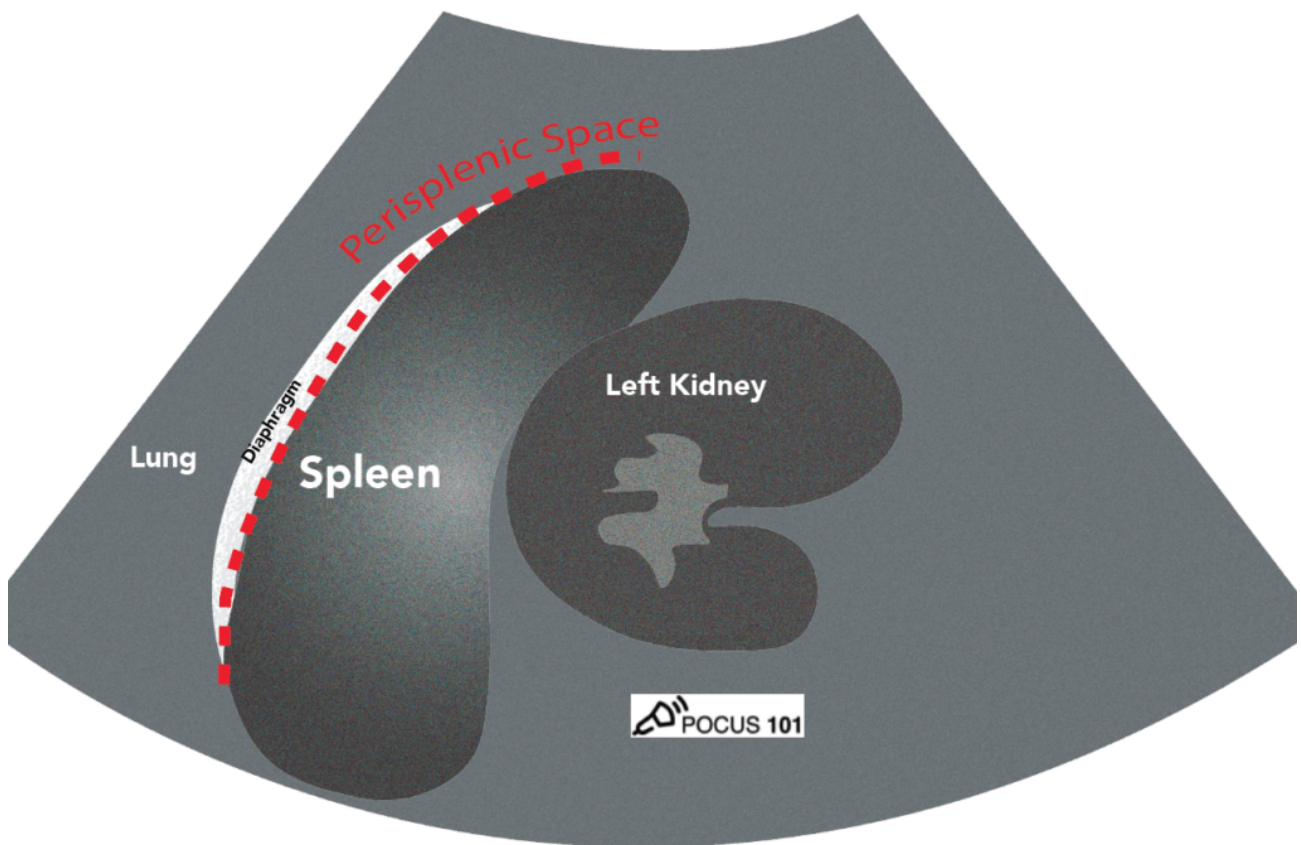


Abdominal Free Fluid in Morison's Pouch

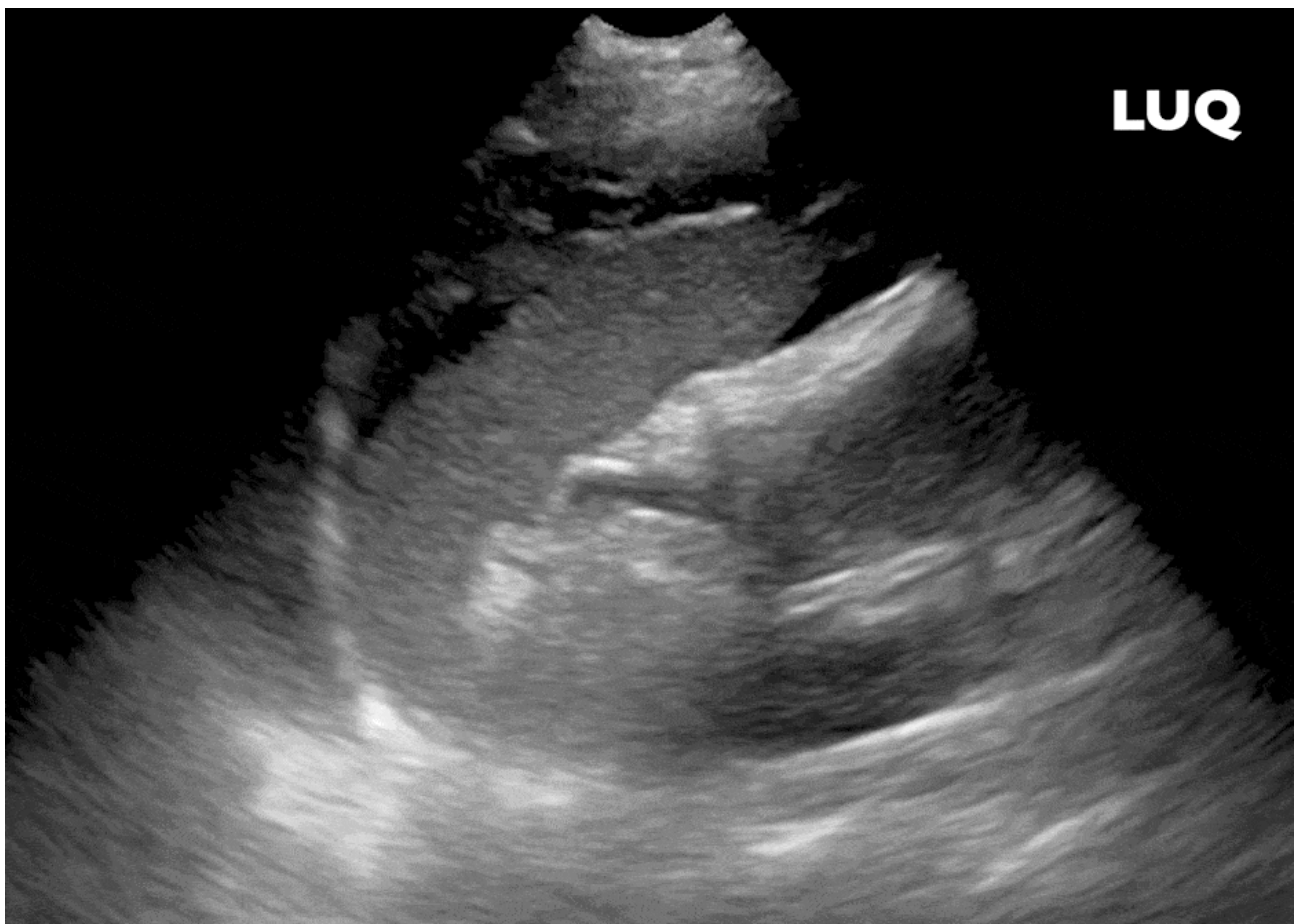
## Left Upper Quadrant – Perisplenic Recess

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Using the technique to scan the spleen described above, you can detect free abdominal fluid in the **perisplenic recess**. You can also check the **costodiaphragmatic recess** from this view.



Perisplenic Space for Free Abdominal Fluid

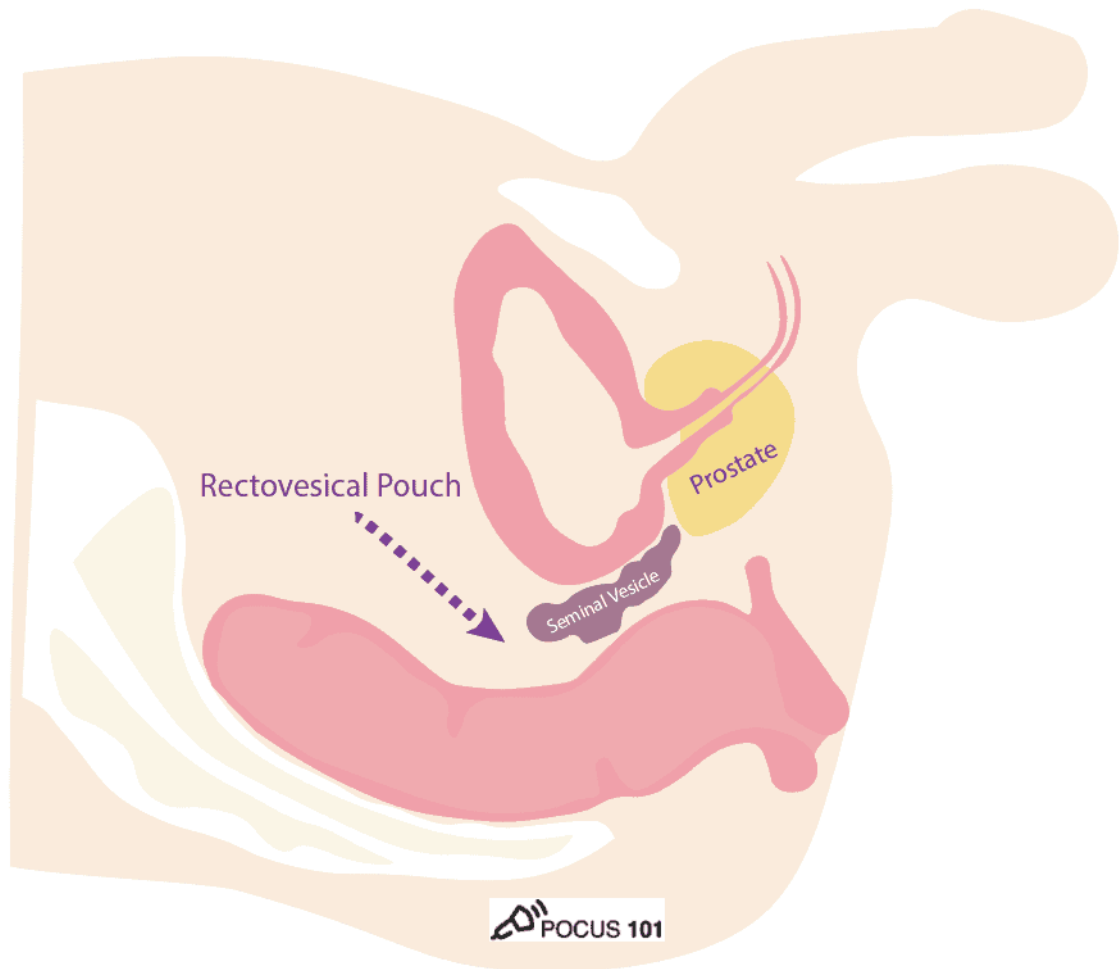


Abdominal Free Fluid in Perisplenic Recess/Space

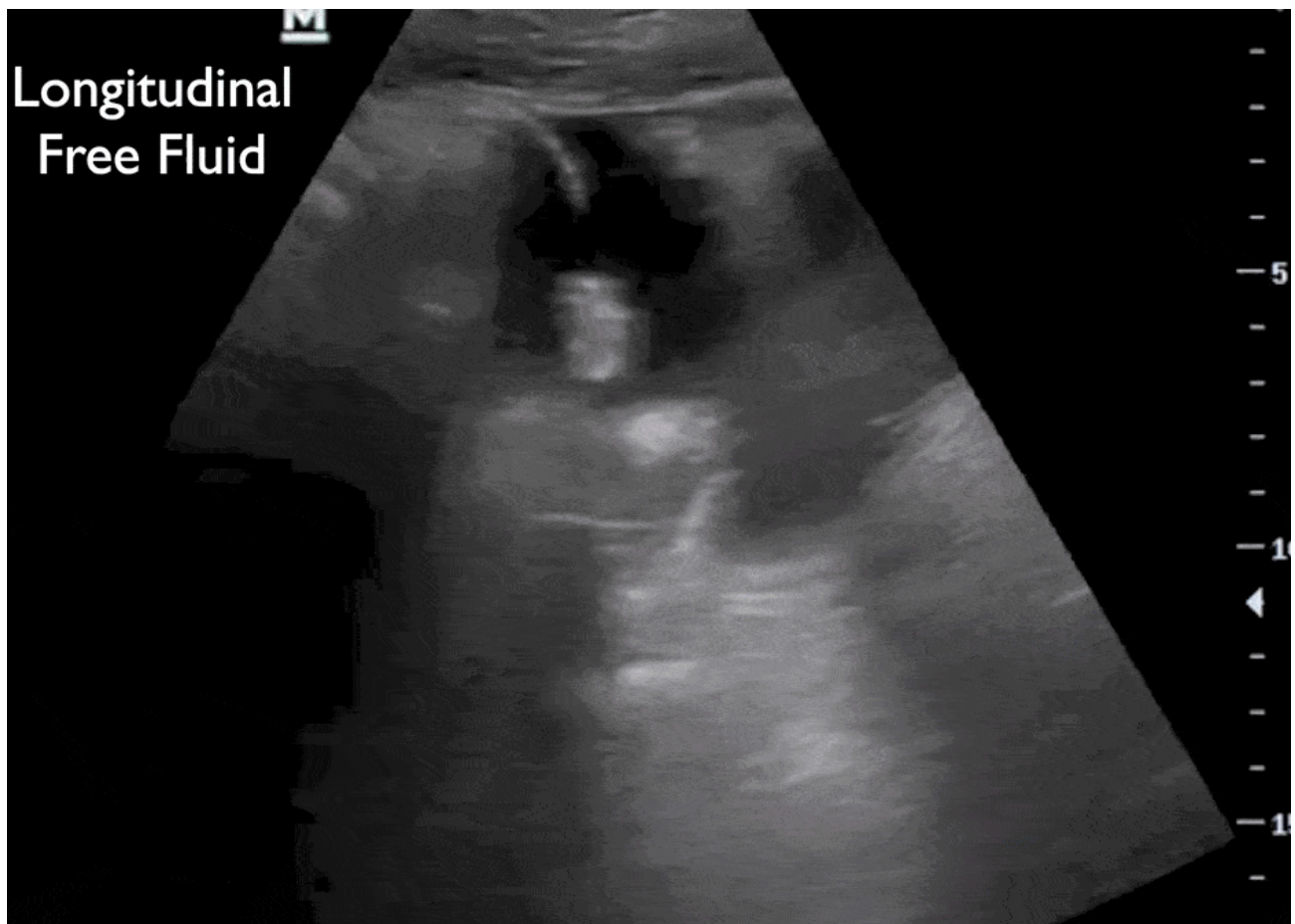
## Male Pelvis – Retrovesical Pouch

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In the male pelvis, you can find free fluid in the rectovesical pouch/space.



Male Pelvis Rectovesical Pouch



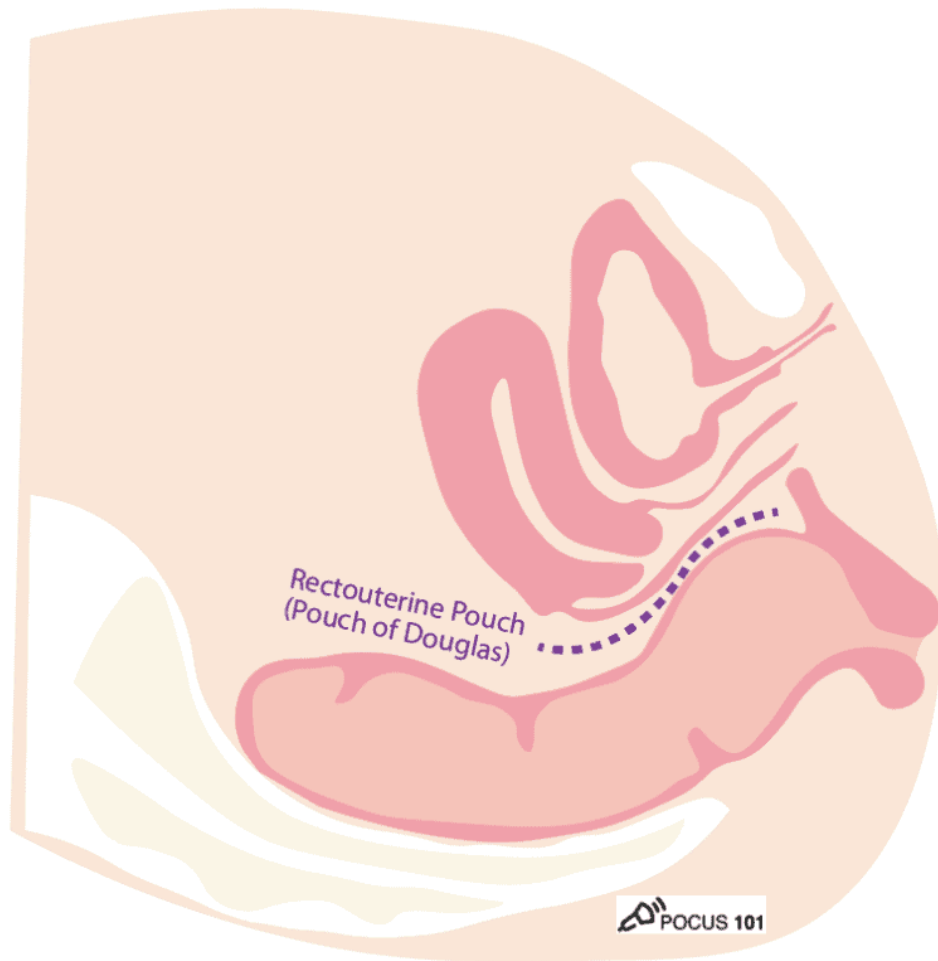
Male – Abdominal Free Fluid in Pelvis

### **Female Pelvis – Pouch of Douglas**

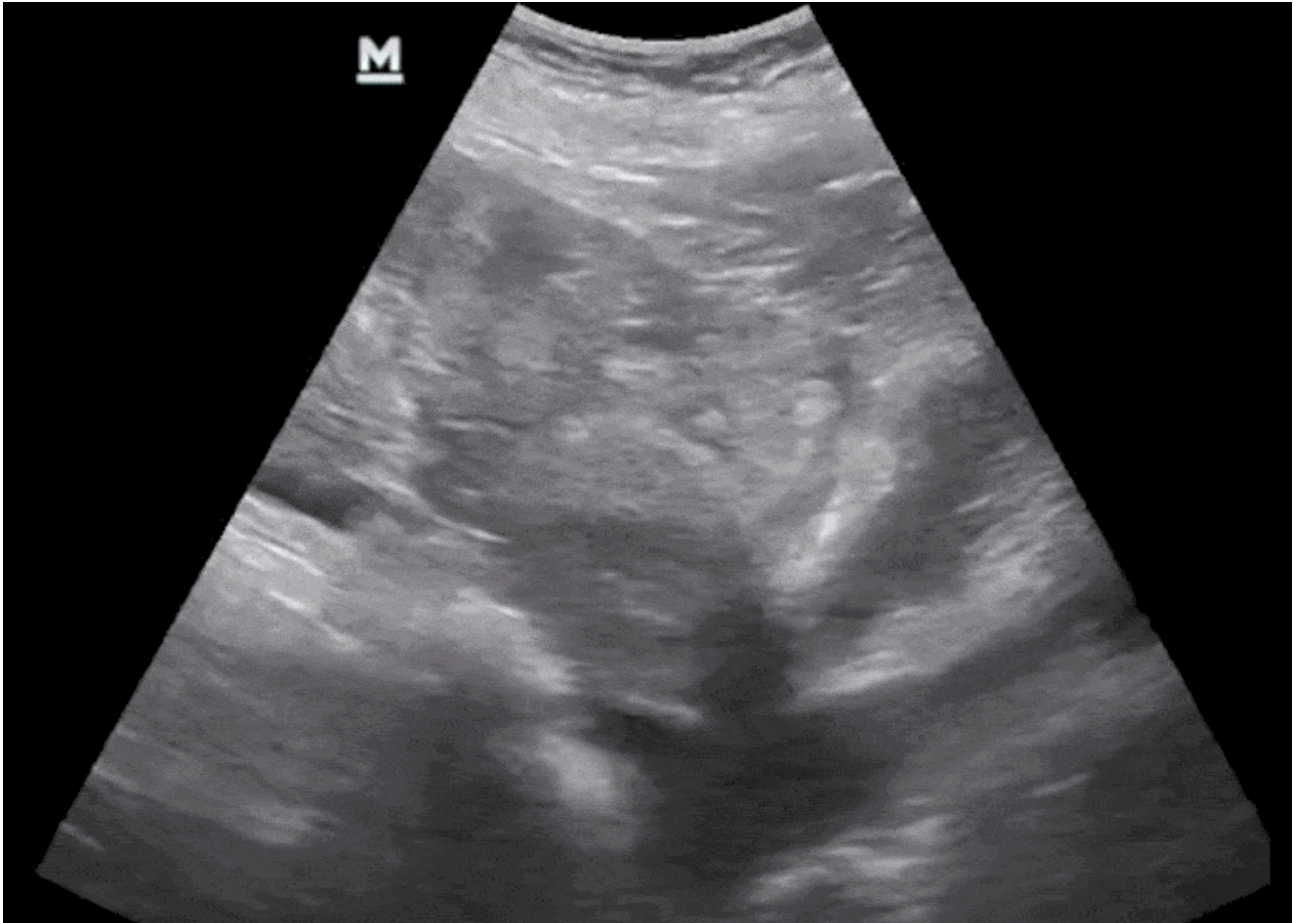
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In the female pelvis, you can find free fluid in the Pouch of Douglas (Rectouterine Pouch).





Female Pelvis Rectovesical Pouch



Female – Abdominal Free Fluid in Pelvis

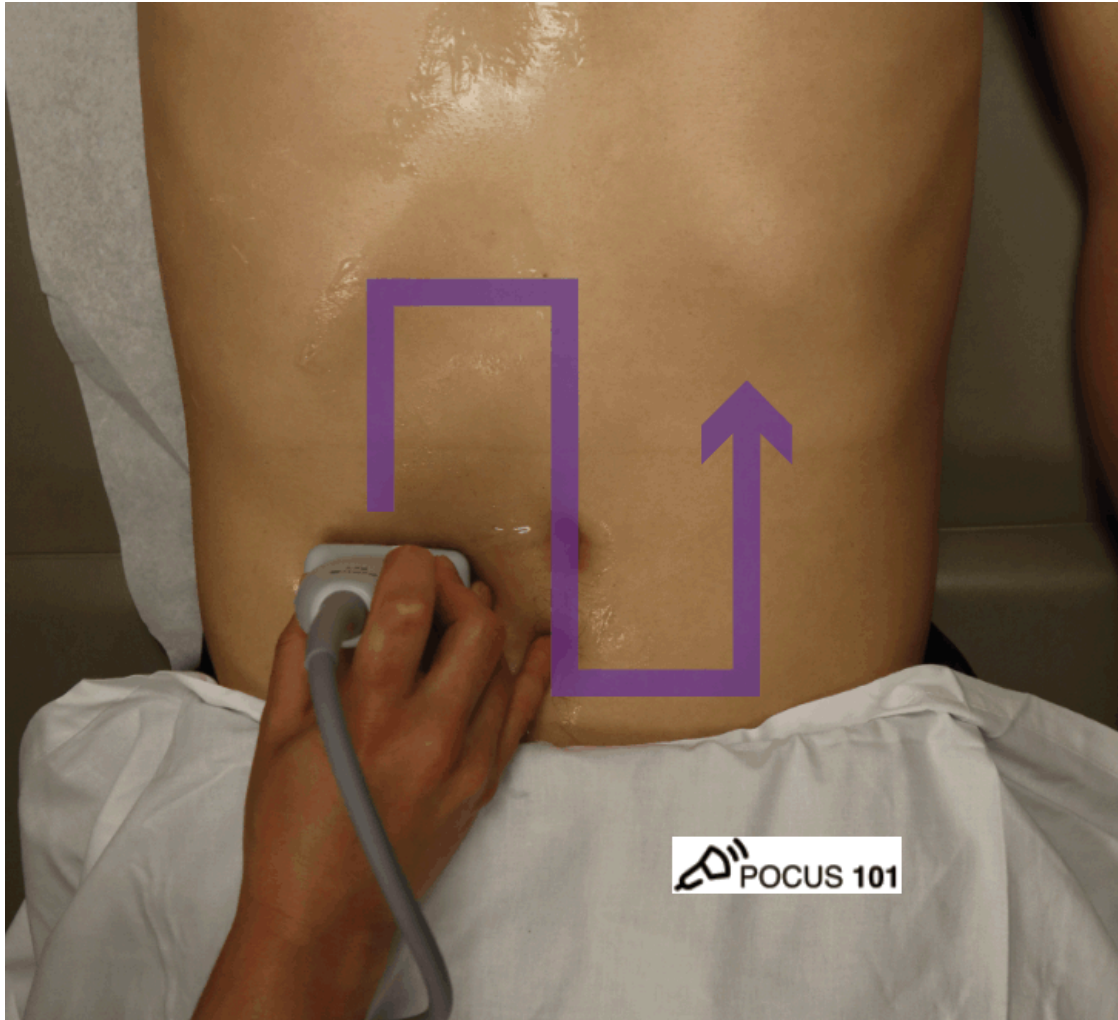
## Small Bowel Obstruction Ultrasound

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Small Bowel Obstruction (SBO) is a common type of mechanical GI obstruction, representing 80% of cases (versus 20% large bowel obstruction). Ultrasound can be used to detect SBO (Taylor).

Classic signs and symptoms for SBO include cramping abdominal pain, and abdominal distension with nausea.

The ultrasound technique for detecting SBO is known as “**mowing the lawn**” or the “lawnmower technique”. Starting in the RLQ and using graded compression, slowly scan the abdominal quadrants as depicted below.



lawnmower technique for Ultrasound for SBO

**Ultrasound findings for small bowel obstruction include:**

- Dilated bowel loops (diameter > 3cm)
- Ineffective or absent peristalsis, which will be seen as a “whirling” of intra-luminal contents
- Collapsed colonic lumen

When 2 of 3 of these findings are present, POCUS has sensitivities and specificities of over 90% to detect SBO (Unlüer et al).



*Dilated bowel loops in the small bowel with haustra.*  
(Unlüer et al)

## Pneumoperitoneum Ultrasound

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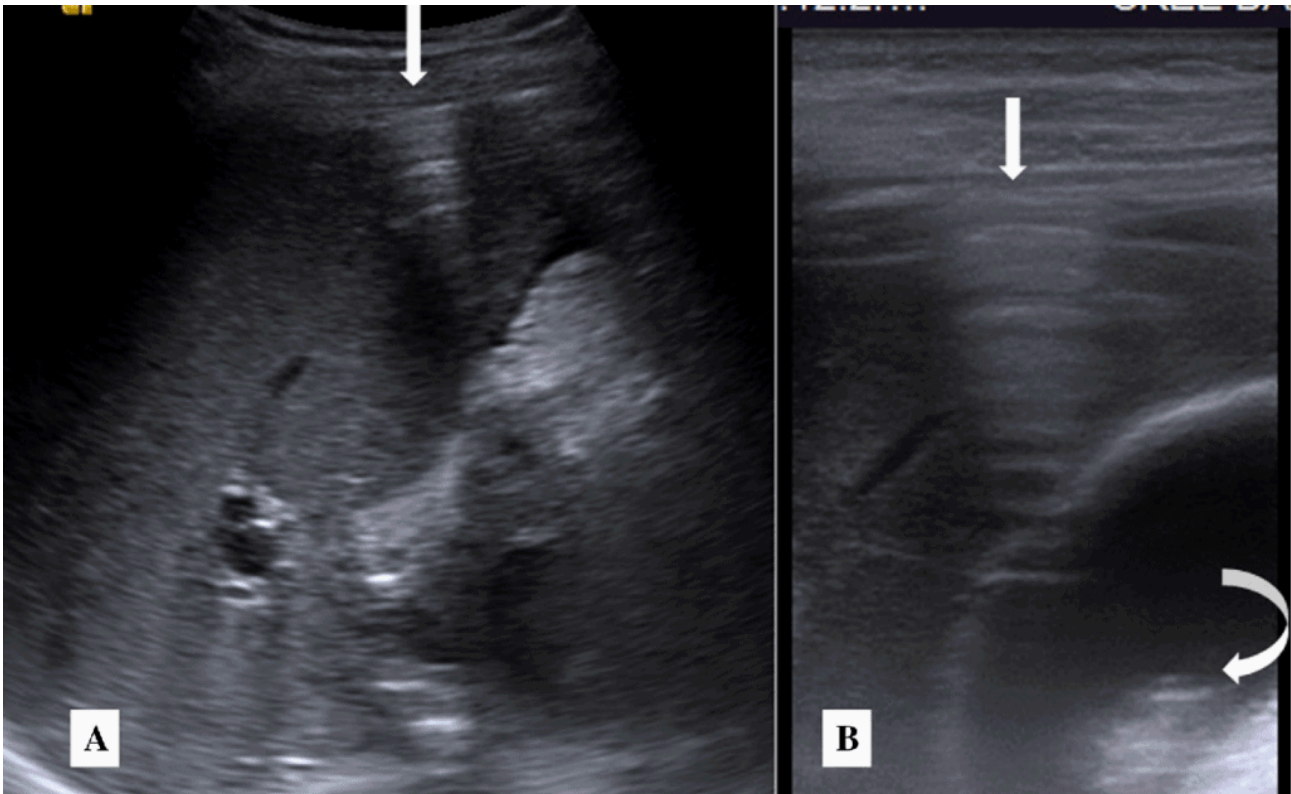
Pneumoperitoneum is free air within the peritoneal cavity. It has various etiologies and is easily confused with several other causes of acute abdominal pain. Etiologies can broadly be classified as traumatic or non-traumatic. Traumatic causes of pneumoperitoneum include iatrogenic injuries or stab wounds. Non-traumatic causes are numerous, including perforated ulcer, infection, erosion, and ischemia.

The most common cause of pneumoperitoneum is the perforation of a hollow viscus, such as in a perforated stomach ulcer (Tanner).

On abdominal ultrasound, the most common finding for pneumoperitoneum is the **Enhanced Peritoneal Stripe Sign (EPSS)**. This is when air within the peritoneal space rises and causes an “echoing” of the usually single, hyperechoic peritoneal stripe that separates the abdominal wall from underlying peritoneal fluid and fluid-filled organs (Indiran).

If there is a large amount of pneumoperitoneum, your image of abdominal organs will be obscured by gas wherever you place your probe.





*Enhanced peritoneal stripe sign (EPSS) seen anterior to the liver in both images (straight arrows), indicating the abnormal presence of air between the liver and the anterior abdominal wall (Indiran).*

## Pediatric Abdominal Ultrasound Applications

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Abdominal ultrasound has numerous pediatric applications. Here, we focus on appendicitis vs mesenteric adenitis, pyloric stenosis, and intussusception.

### Appendicitis vs Mesenteric Adenitis

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Our technique for detecting appendicitis is described in the section above.

An important differential diagnosis in appendicitis includes **mesenteric adenitis**. In contrast to appendicitis, ultrasound findings for mesenteric adenitis will be the absence of a visibly enlarged appendix, and the presence of enlarged mesenteric nodes, as pictured below.



*Enlarged mesenteric lymph nodes.*

Case courtesy of Dr Chris O'Donnell, [Radiopaedia.org](http://Radiopaedia.org).

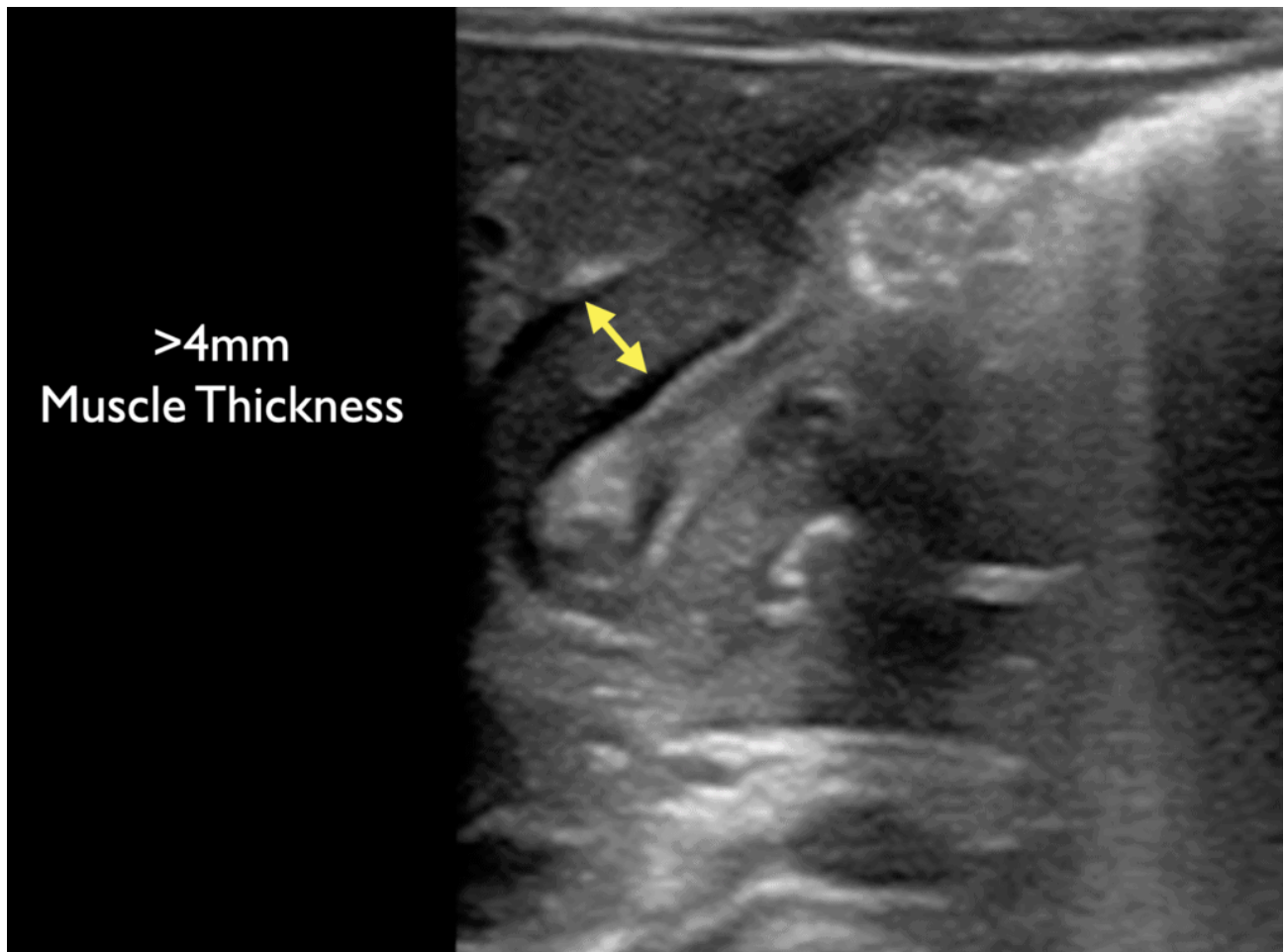
From the case [rID: 18049](#)

## Pyloric Stenosis Ultrasound

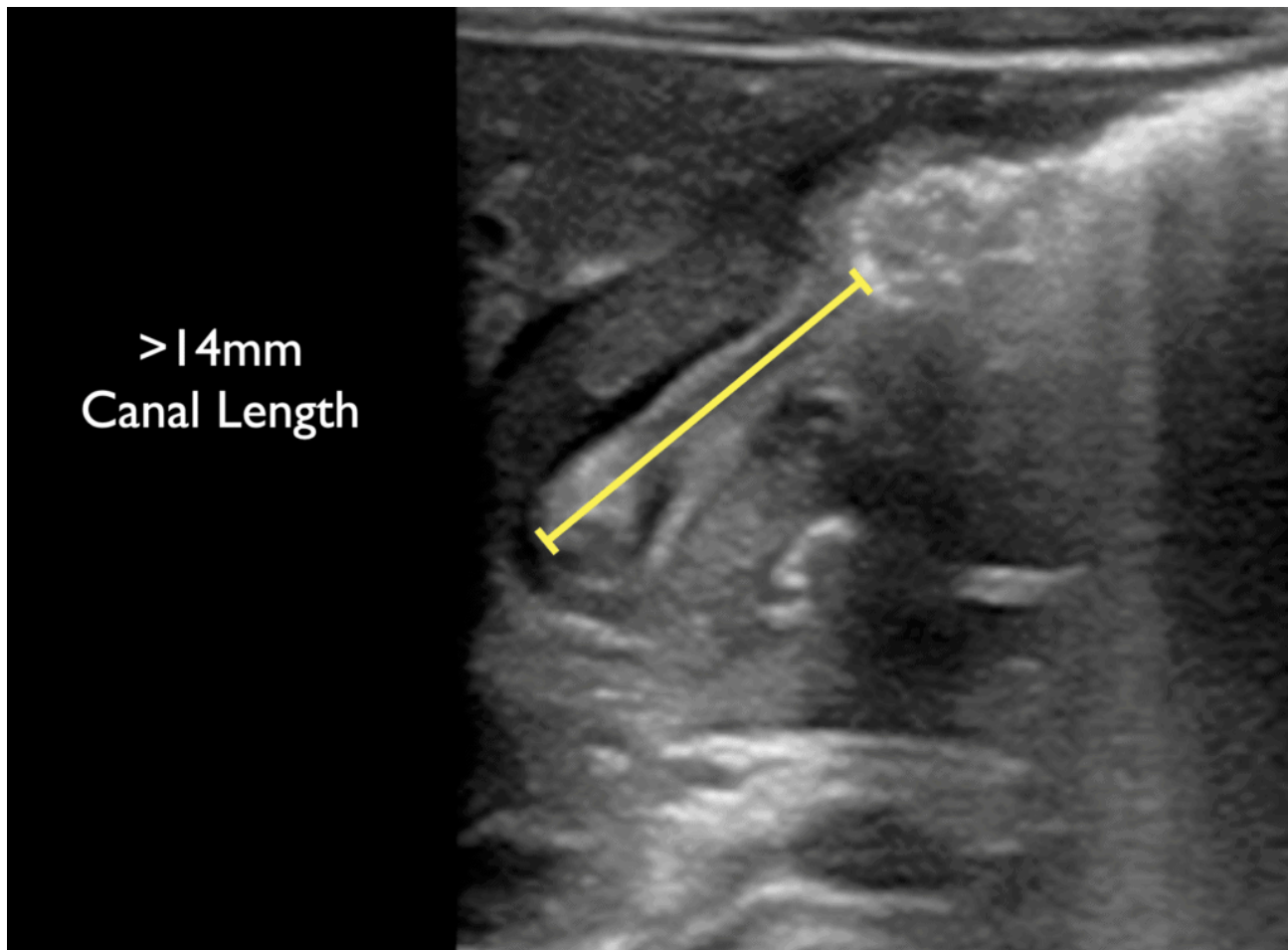
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Pyloric stenosis refers to idiopathic, hypertrophic thickening of the pyloric musculature, which progresses to gastric outlet obstruction. Symptoms typically present at 2-8 weeks of age. Classically, this will feature a neonate with non-bilious, projectile vomiting. You may also appreciate an olive-sized mass in the right upper quadrant on physical exam.

On ultrasound, the diagnosis of pyloric stenosis is based on 2 measurements: **a pyloric muscle diameter > 4mm**, and a **pyloric canal length > 14mm**.



Pyloric Stenosis on Ultrasound >4mm Muscle Thickness



Pyloric Stenosis on Ultrasound >14mm Canal Length

## Intussusception Ultrasound

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Intussusception occurs when a segment of bowel “telescopes” onto another segment, or onto itself, because of peristalsis. It is a common cause of acute abdomen in children, the most common cause of intestinal obstruction in children, and has significant complications, such as bowel necrosis, making its timely detection imperative (Edwards).

The classic triad of intussusception is intermittent abdominal pain, vomiting, and the presence of a right upper quadrant mass. When paired with the presence of gross or occult bleeding on rectal exam, this has a high positive predictive value for intussusception.

POCUS has high sensitivity and specificity for detection of intussusception (97.9% and 97.8%, respectively), as well as a high negative predictive value (99.7%; Edwards).

The most common ultrasound findings for intussusception are:

- **Target sign:** inflamed bands of hyperechoic mucosa and relatively hypoechoic submucosa are visible.
- **Pseudokidney sign:** fat-containing mesentery is dragged into the intussusception, resembling the kidney hilum. The resultant longitudinal appearance is of a small kidney.





*The Target Sign.*

Case courtesy of Dr Behrang Amini ,  
[Radiopaedia.org](http://Radiopaedia.org). From the case [rID:](#)  
[3456](#)



*The pseudo kidney sign.*

Case courtesy of Dr T Menezes, [Radiopaedia.org](http://Radiopaedia.org). From the case [rID:](#)  
[33056](#)

## References

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1. Noble, V. E., & Nelson, B. (2014). **Manual of Emergency and Critical Care Ultrasound**. Cambridge, UK, MA: Cambridge University Press.
2. Rumack, Carol M., and Levine, Deborah. **Diagnostic Ultrasound E-Book**. United Kingdom, Elsevier Health Sciences, 2017.
3. Arger, Peter, & Benson, Carol. **Ultrasound, a practical approach to clinical problems**. Thieme Publishing Group. (2008) ISBN:3131168323.
4. Edlin BR, Eckhardt BJ, Shu MA, Holmberg SD, Swan T. **Toward a more accurate estimate of the prevalence of hepatitis C in the United States**. *Hepatology*. 2015 Nov;62(5):1353-63. doi: 10.1002/hep.27978. Epub 2015 Aug 25. PMID: 26171595; PMCID: PMC4751870.
5. Office of Infectious Disease and HIV/AIDS Policy (OIDP). (2016, June 7). **Viral Hepatitis in the United States: Data and Trends**. HHS.Gov. <https://www.hhs.gov/hepatitis/learn-about-viral-hepatitis/data-and-trends/index.html>
6. Scaglione S, Kliethermes S, Cao G, Shoham D, Durazo R, Luke A, Volk ML. **The Epidemiology of Cirrhosis in the United States: A Population-based Study**. *J Clin Gastroenterol*. 2015 Sep;49(8):690-6. doi: 10.1097/MCG.0000000000000208. PMID: 25291348.
7. Macnaught F, Campbell-Rogers N. **The liver: how we do it**. *Australas J Ultrasound Med*. 2009;12(3):44-47. doi:10.1002/j.2205-0140.2009.tb00061.x
8. **Normal Liver Ultrasound How To**. (2020). Retrieved August 12, 2020, from <https://www.ultrasoundpaedia.com>
9. Heller MT, Tublin ME. **The role of ultrasonography in the evaluation of diffuse liver disease**. *Radiol Clin North Am*. 2014;52(6):1163-1175. doi:10.1016/j.rcl.2014.07.013
10. Tchelepi H, Ralls PW, Radin R, Grant E. **Sonography of diffuse liver disease**. *J Ultrasound Med*. 2002;21(9):1023-1034. doi:10.7863/jum.2002.21.9.1023
11. Lurie, Yoav et al. **Non-invasive diagnosis of liver fibrosis and cirrhosis**. *World journal of gastroenterology* vol. 21,41 (2015): 11567-83. doi:10.3748/wjg.v21.i41.11567
12. Khov, N., Sharma, A., Riley, T. (2014). **Bedside ultrasound in the diagnosis of nonalcoholic fatty liver disease** *World Journal of Gastroenterology* 20(22), 6821-6825.
13. Case courtesy of Dr Bruno Di Muzio, [Radiopaedia.org](https://radiopaedia.org). From the case [rID: 80805](#)
14. Joshi, G., Crawford, K., Hanna, T., Herr, K., Dahiya, N., Menias, C. (2018). **US of Right Upper Quadrant Pain in the Emergency Department: Diagnosing beyond Gallbladder and Biliary Disease** [RadioGraphics 38\(3\), 766-793](#).
15. Pinto, A., Reginelli, A., Cagini, L. *et al.* **Accuracy of ultrasonography in the diagnosis of acute calculous cholecystitis: review of the literature**. *Crit Ultrasound J* 5, S11 (2013). <https://0-doi-org.catalog.illu.edu/10.1186/2036-7902-5-S1-S11>

16. Revzin, M., Scoutt, L., Garner, J., Moore, C. (2017). **Right Upper Quadrant Pain: Ultrasound First!** *Journal of Ultrasound in Medicine* 36(10), 1975-1985  
<https://dx.doi.org/10.1002/jum.14274>
17. Liao, K., Teh, C., & Serrablo, A. (2014). **Management of acute cholecystitis and acute cholangitis in emergency setting**, *Open Medicine*, 9(3), 357-369.  
doi: <https://doi.org/10.2478/s11536-013-0301-3>
18. Bree, RL. **Further observations on the usefulness of the sonographic Murphy sign in the evaluation of suspected acute cholecystitis.** *J Clin Ultrasound*. 1995 Mar-Apr;23(3):169-72. doi: 10.1002/jcu.1870230304. PMID: 7730462.
19. Bell, D., & Sorrentino, S. (2012). **Gallstones: Radiology Reference Article.** Retrieved October 05, 2020
20. Kothari, S. N., Obinwanne, K. M., Baker, M. T., Mathiason, M. A., & Kallies, K. J. (2013). **A Prospective, Blinded Comparison of Laparoscopic Ultrasound with Transabdominal Ultrasound for the Detection of Gallbladder Pathology in Morbidly Obese Patients.** *Journal of the American College of Surgeons*, 216(6), 1057-1062. doi:10.1016/j.jamcollsurg.2013.02.009
21. Hartung, M., & Gaillard, F. (2010). **Choledocholithiasis: Radiology Reference Article.** Retrieved October 05, 2020
22. Mostbeck, G., Adam, E. J., Nielsen, M. B., Claudon, M., Clevert, D., Nicolau, C., Nyhsen, C., & Owens, C. M. (2016). **How to diagnose acute appendicitis: ultrasound first.** *Insights into imaging*, 7(2), 255–263. <https://doi.org/10.1007/s13244-016-0469-6>
23. Al-Khayal KA, Al-Omran MA. **Computed tomography and ultrasonography in the diagnosis of equivocal acute appendicitis. A meta-analysis.** *Saudi Med J*. 2007 Feb;28(2):173-80. PMID: 17268692.
24. Unlüer EE1, Yavaş O, Eroğlu O, et al. **Ultrasonography by emergency medicine and radiology residents for the diagnosis of small bowel obstruction.** *Eur J Emerg Med*. 2010 Oct;17(5):260-4.)
25. Taylor MR, Lalani N. **Adult small bowel obstruction.** *Acad Emerg Med*. 2013 Jun;20(6):528-44. doi: 10.1111/acem.12150. PMID: 23758299.
26. Edwards, E.A., Pigg, N., Courtier, J. et al. **Intussusception: past, present and future.** *Pediatr Radiol* **47**, 1101–1108 (2017). <https://doi.org/10.1007/s00247-017-3878-x>
27. Tanner TN, Hall BR, Oran J. **Pneumoperitoneum.** *Surg Clin North Am*. 2018 Oct;98(5):915-932. doi: 10.1016/j.suc.2018.06.004. PMID: 30243453.
28. Indiran, V., Kumar, R. V., & Jefferson, B. (2018, May 3). **Enhanced peritoneal stripe sign.** *Abdominal Radiology*, 43, 3518-3519.